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THE COORDINATION CLIFF: HOW CONTEXT COMPRESSION BREAKS MULTI-FRAGMENT LLM WORKFLOWS

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ABSTRACT

Every token a multi-step LLM workflow reads costs energy, latency, and money, so context compression is the obvious lever. The prompt-compression literature is mature, but it evaluates compressors on single-fragment question answering, whereas production workflows must combine information spread across several fragments, each compressed on its own. Whether a compressor that preserves single-fragment quality also preserves that harder, multi-fragment property had never been tested.

It does not. On a new 150-instance benchmark spanning cross-document fact aggregation, constraint-satisfaction planning, and multi-step retrieval, single-fragment retention quality fails to predict multi-fragment coordination success for every training-free compressor tested, and the two measures even disagree on how to rank compressors, so the single-agent benchmarks that dominate the field mis-rank them for multi-fragment use. The failure is abrupt rather than gradual: a sharp *coordination cliff* at which success collapses once compression strips the task-critical tokens a planner must combine. A compounding-error model explains why the transition is sharp and recovers its position to first order. The cliff is a property of the compressor and the task, not the planner: within a calibrated regime its position shows no detected shift from a small local model to a frontier model an order of magnitude larger, even across a change of architecture family.

Turning these findings into a deployable system is the thesis's principal engineering contribution: a compression-aware memory bus that sits between the data sources and the planner and compresses every fragment in transit, converting the token savings directly into lower inference cost. Its compression layer is governed by the measurements rather than by guesswork: a cliff-aware wrapper (CAAC) backs each fragment off to the most aggressive ratio that still keeps it on the safe side of the cliff, while a summary-level disclosure budget caps how much a downstream reader can recover from a compressed confidential fragment. The reference implementation is a policy-enforcing service with a tamper-evident audit log, slated for deployment in the production workflows of the industry partner, TalentAdore, where token cost dominates operating expense.

Keywords: context compression, large language models, coordination cliff, multi-fragment workflows, memory bus, inference disclosure, agentic memory, retrieval-augmented generation

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TIIVISTELMÄ

Suuria kielimalleja hyödyntävät monivaiheiset työnkulut maksavat jokaisesta lukemastaan tokenista, joten kontekstin tiivistäminen on luonteva keino karsia kustannuksia. Kehotteentiivistämisen kirjallisuus on kypsää, mutta se arvioi tiivistäjiä yhden fragmentin kysymys–vastaus-tehtävillä, vaikka tuotannon työkuormat joutuvat yhdistämään tietoa useasta erikseen tiivistetystä fragmentista. Sitä, säilyttääkö yhden fragmentin laadun säilyttävä tiivistäjä myös tämän vaikeamman, monifragmenttisen ominaisuuden, ei ollut aiemmin tutkittu.

Ei säilytä. Uudessa 150 instanssin vertailuaineistossa, joka kattaa faktojen yhdistämisen dokumenttien välillä, rajoiteperustaisen suunnittelun ja monivaiheisen tiedonhaun, yhden fragmentin tiedon säilyminen ei ennusta monifragmenttisen koordinaation onnistumista yhdelläkään testatulla koulutusvapaalla tiivistäjällä, ja mittarit jopa eroavat siitä, miten tiivistäjät tulisi asettaa paremmuusjärjestykseen. Näin alaa hallitsevat yhden agentin vertailuaineistot asettavat tiivistäjät väärään järjestykseen monifragmenttista käyttöä ajatellen. Romahdus on äkillinen, ei asteittainen: terävä koordinaation jyrkänne, jossa onnistuminen romahtaa heti, kun tiivistäminen karsii ne tehtävän kannalta kriittiset tokenit, jotka suunnittelijan on yhdistettävä. Kumuloituvan virheen malli selittää, miksi siirtymä on jyrkkä, ja palauttaa sen sijainnin ensimmäisen kertaluvun tarkkuudella. Jyrkänne on tiivistäjän ja tehtävän, ei suunnittelijan, ominaisuus: kalibroidulla alueella sen sijainnissa ei havaita siirtymää pienestä paikallisesta mallista suuruusluokkaa suurempaan huipputason malliin, ei myöskään arkkitehtuuriperheen vaihtuessa.

Näiden havaintojen muuntaminen käyttöönotettavaksi järjestelmäksi on opinnäytetyön keskeinen tekninen kontribuutio: tiivistystietoinen muistiväylä, joka sijaitsee tietolähteiden ja suunnittelijan välissä, tiivistää jokaisen fragmentin matkalla ja muuntaa tokensäästöt suoraan pienemmäksi päättelykustannukseksi. Sen tiivistyskerrosta ohjaavat arvausten sijaan mittaukset: jyrkännetietoinen kääre (CAAC) perääntyy kunkin fragmentin kohdalla aggressiivisimpaan suhteeseen, joka pitää fragmentin jyrkänteen turvallisella puolella, kun taas yhteenvetotason päättelyilmaisubudjetti rajoittaa sitä, kuinka paljon loppulukija voi palauttaa tiivistetystä luottamuksellisesta fragmentista. Viitetoteutus on käytäntöä valvova palvelu, jossa on väärentämisen paljastava tarkastusloki, ja se on määrä ottaa käyttöön teollisuuskumppani TalentAdoren tuotantotyönkuluissa, joissa token-kustannus hallitsee käyttömenoja.

Avainsanat: kontekstin tiivistäminen, suuret kielimallit, koordinaatiojyrkänne, monifragmenttiset työnkulut, muistiväylä, päättelyilmaisu, agenttinen muisti, hakuparannettu generointi

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FOREWORD

This thesis was carried out at the Future Computing Group, CSE Research Unit, Faculty of Information Technology and Electrical Engineering, University of Oulu, in collaboration with TalentAdore Ltd. The work was supervised academically by Lauri Lovén, with industrial supervision from Asim Nadeem, Oskari Valkama, and Saku Valkama at TalentAdore.

My deepest thanks go to Lauri, whose constant and unwavering support and patience carried this work from a vague idea to a finished thesis. He gave me the latitude to chase an unfamiliar empirical question (how coordination under compression behaves at institutional scale) and the steady, patient guidance to keep the scope tight whenever it threatened to sprawl; I could not have asked for a better supervisor. I am just as grateful to Asim Nadeem, Oskari Valkama, and Saku Valkama at TalentAdore for their support throughout the thesis, and for sharing TalentAdore's production cost structure, which anchored the cost-per-workflow arm of the evaluation in a real industrial setting.

The thesis was prepared on a single Apple M4 Pro 48 GB workstation with an RTX 5090 32 GB GPU host on the Tailscale mesh for the compression precomputation and the model-scaling and frontier-API sweeps. The compute envelope, the reproducibility package, and the reference implementation accompanying the thesis are described in Appendix 7, Section D.

This work was supported by the Technology Industries of Finland Centennial Foundation (Teknologiateollisuuden 100-vuotissäätiö) under grant number 6193.

On the use of AI tools. I leaned on generative-AI tools throughout this thesis (mainly Anthropic's Claude through the Claude Code command-line interface), and they were a genuine productivity boost: kicking ideas around, writing and debugging the experiment and analysis code, and tightening the prose. I used them as a fast assistant rather than an authority: the hypotheses, the experimental results, and their interpretation are my own, checked against the on-disk data, and I take responsibility for everything in this thesis. Used this way, the tools made the work faster and the writing cleaner without standing in for the thinking.

Helsinki, June 2026

Syed Abdullah Hassan

Teknologiateollisuuden
100-vuotissäätiö

LIST OF ABBREVIATIONS AND SYMBOLS

ACL	access control list
API	application programming interface
AUC	area under the curve
BCa	bias-corrected and accelerated (bootstrap CI)
CAAC	cliff-aware adaptive compression
CI	confidence interval
C1 . . . C4	contributions one through four of this thesis
CSE	Computer Science and Engineering
CTR	critical-token-recall (per-family task-relevant token recall)
EM	exact match (QA metric)
EUR	euro (currency)
FAISS	Facebook AI Similarity Search
FCG	Future Computing Group
H1 . . . H6	hypotheses one through six
HNSW	hierarchical navigable small world (vector index)
HTTP	hypertext transfer protocol
ICAE	in-context autoencoder
ITEE	Information Technology and Electrical Engineering
JSON	JavaScript Object Notation
KV cache	key–value cache (transformer attention)
LLM	large language model
LOO	leave-one-(family-)out cross-validation
LoRA	low-rank adaptation
LRU	least recently used
MLX	Apple’s native ML framework for Apple Silicon
MPS	Metal Performance Shaders (Apple PyTorch backend)
NIST AI RMF	NIST AI Risk Management Framework
OAuth	open authorisation protocol
OTel	OpenTelemetry
P1 . . . P3	retrieval–compression pipeline variants one through three
pp	percentage points
QA	question answering
RAG	retrieval-augmented generation
REST	representational state transfer
SHA-256	secure hash algorithm (256-bit)
SQL	structured query language
SQLite	a lightweight, embedded SQL database
SSE	server-sent events
TF-IDF	term frequency–inverse document frequency
TOST	two one-sided tests (statistical equivalence test)
TTL	time-to-live
UUID	universally unique identifier
WAL	write-ahead log

Symbols of the compounding-error model (Chapter 4, Section 4.6):

r	nominal compression ratio (source / output token count)
$q(r)$	per-round critical-token-recall at ratio r
θ_q	cliff-recall threshold (per-family), the minimum recall required for coordination success
θ_{info}	task information-density estimate (per-task, AUC-based; distinct from θ_q)
τ^*	cliff position, the ratio at which coordination success drops below the cliff threshold
p_0	baseline coordination success at $r = 1$ (no-compression baseline)
N	number of sequential compression passes (always 1 in the empirical chapters of this thesis)

Statistical symbols:

ρ	Spearman's rank correlation coefficient
p	two-sided p-value (Holm-corrected within hypothesis family unless noted)
n_{boot}	bootstrap resample count (default 10,000 for H1/H2/H3/H4 paths; 500 for the frontier-τ^* and θ_q pipelines; BCa interval throughout)

1. INTRODUCTION

1.1. Motivation: Every Token Has a Cost

Large language models pay for every token they read. The cost shows up in three places: the energy and water to read the prompt and generate the reply (the prefill and decoding passes); the latency the user waits through; and the dollars the operator pays per million tokens, through an API or amortised over on-premises hardware. Peer-reviewed measurements of this inference-side cost are recent but consistent. Luccioni *et al.*'s 2024 *Power Hungry Processing* study reports per-inference Watt-hour numbers for a broad set of open models, about 0,1 Watt-hours per inference for a 6–7B-class open model [1]; Samsi *et al.* [2] show the energy of LLM inference scales with the number of tokens processed; and Pope *et al.* [3] quantify how serving cost grows with sequence length once the prefill dominates. Patterson *et al.* [4] established the training-time carbon-and-cost accounting that these inference-time studies extend. When the workload is a multi-step agentic system that issues thousands of internal inference calls to answer one user request, those per-token costs compound.

A growing fraction of the tokens a deployed LLM reads do not change the answer. They are conversational filler ("Hello", "Thanks"), repeated system prompts that scaffold every call, restated history that re-anchors a long conversation, boilerplate disclaimers, and framing an attentive reader could skip. Because the measurements above tie cost directly to the number of tokens processed [1–3], each such token still turns into energy, latency, and operating expense whether or not it changes the answer. As a consistent industry anecdote, OpenAI's compute bill for processing user politeness alone was informally estimated at "tens of millions of dollars" a year in 2025 [5]; it is corroborating colour, not evidence.

Context compression is the natural lever. Per-token compute and cost scale roughly linearly in sequence length once the prefill is past the GPU memory-bound regime, so halving the input context roughly halves the cost. The single-agent prompt-compression literature has matured fast over the past three years [6–11]. This thesis measures what that literature has left untested: whether compression that preserves single-fragment quality (the single-agent question-answering metric it optimises) also preserves the harder property a planner needs, combining information spread across several fragments into one correct answer.

1.2. The Multi-Fragment Workflow

This thesis studies the *multi-fragment LLM workflow*: a task whose answer cannot be read out of any single fragment and instead requires combining information across two or more fragments, each of which may be compressed on its own before the planner sees it. Three examples recur: cross-document fact aggregation (sum eight numbers reported by eight institutional systems), constraint-satisfaction planning (assign N sub-tasks to K capacity-bounded workers from specifications spread across fragments), and multi-step retrieval (chain references through a sequence of documents to a leaf answer); the C1 benchmark of Chapter 3 builds 50 instances of each. This setting is not a corner case. The FCG group at the University of Oulu's flagship use case, *Vignette 3.7 – Period-end project reporting* [12], gathers reporting data across eight university systems via roughly eight specialised agents; the internal FCG analysis estimates an 80% cloud-payload reduction at about EUR 2,15 per report. TalentAdore, the industry partner of this thesis, runs production workloads whose operating expense is dominated by token cost. Anthropic's report on their multi-agent research system finds token usage explains about 80% of the performance variance they observe [13], and the companion context-management report describes an 84% token reduction on a 100-turn web-search evaluation [14]. These are industry observations on industry workflows; the master's-scale problem is the controlled cross-compressor, cross-task, cross-architecture measurement that turns that urgency into a technical question.

The technical question deliberately excludes the multi-round agent negotiation that the "multi-agent" framing of the FCG and Anthropic systems implies. The empirical evaluation uses one of two planners: a deterministic regex information-extraction solver, or a single LLM call with all (compressed) fragments visible. An AutoGen v0.4 multi-round backend ships in the source tree (`src/m6/orchestrator/` [15]) but is not used in any reported result. The reason is attribution: round-to-round LLM variance dominates the compression signal at the C1 ratios swept, so cleanly attributing a coordination drop to compression rather than to agent variance requires isolating the compressor on the critical path, which a single-LLM-call or deterministic planner does. The memory bus is *designed for* multi-agent integration; the experimental scope is multi-fragment task solvability, restated and calibrated in Chapter 3 Section 3.8.

Operational definition of "coordination success." Because the title and the coinage *coordination cliff* invite a multi-agent reading the experiments do not test, the term is fixed here and used uniformly thereafter. **Coordination success is the binary indicator that the planner recovers the correct answer to a multi-fragment task whose answer cannot be read out of any single fragment alone**: a measurement of *multi-fragment solvability under compression*, not of multi-round agent negotiation, message-passing efficiency, or role allocation under disagreement. Every later use of "coordination" refers to this quantity.

1.3. Problem Statement

How does context compression affect multi-fragment coordination quality across compressors, task structures, and planner scales, and when and how can a memory-bus service exploit that understanding to compress safely, predictably, and privately?

The question subdivides into four sub-questions, one per contribution (C1–C4), each answered end-to-end by an artefact and operationalised by one or more of the falsifiable hypotheses (H1–H6) that structure Chapter 4.

1.3.1. Research Questions and Their Mapping to Hypotheses

The four research questions and their hypothesis mappings are:

RQ1 (metric). Does a coordination-quality metric on a multi-fragment workflow behave differently from single-agent question-answering accuracy? (operationalised by **H1**).

RQ2 (characterisation). Does compression induce a coordination cliff, and what determines its position? (operationalised by **H2** (existence and sharpness), **H5/Corollary 1** (invariance to planner scale), and **H6/Corollary 2** (scaling with task information density)).

RQ3 (deployment). How should compression be combined with retrieval, and what is the cost trade-off across pipeline architectures? (operationalised by **H3**).

RQ4 (governance). Can compression be measured as a privacy lever and exposed as a deployment-time enforcement mechanism by a memory-bus service? (operationalised by **H4** and the memory-bus integration of Chapter 4).

1.4. Contributions

Headline. The thesis's single most-citable result is the H1 finding (the methodological backbone of C2): on the multi-fragment benchmark shipped as C1, the change in single-fragment source-retention F1 under compression does not positively predict the change in multi-fragment coordination success for *any* of the four compressors tested. The single-agent benchmarks that dominate the prompt-compression literature (LLMLingua-2, LongLLMLingua, LongBench, RULER) therefore systematically mis-rank compressors for multi-fragment deployment. The thesis turns that finding into its principal engineering deliverable: a *compression-aware memory bus* (C4) whose compression layer is tuned by the cliff measurements rather than by guesswork, built to cut inference cost in production.

C1: A reproducible multi-fragment coordination benchmark. *The C1 benchmark* ships 150 instances across three workload families (cross-document fact aggregation, constraint-satisfaction planning, multi-step retrieval), with synthetic provenance and access-control tag distributions, four coordination-quality metrics, and single-command regeneration from a fixed seed. It is the substrate on which the cliff characterisation, the model-scaling result, and the disclosure measurement all run; releasing it lets other groups extend the characterisation to new compressors and planner regimes.

C2: Empirical characterisation of the coordination cliff, a compounding-error model that explains it, and two structural corollaries. *The cliff exists and is sharp; its position shows no detectable shift across the heterogeneous planners we could test within a defined calibrated regime; and its shape varies systematically with task information density.* The empirical work covers four training-free compressors (LLMLingua-2 [8], a Phi-3-Mini extractive span selector, an instruction-aware filter, and a truncation baseline) at ten ratios across three families and three local architecture families (Qwen-2.5-1.5B, Phi-3-Mini-3.8B, Llama-3.1-8B), with a frontier consistency arm (Qwen-2.5-72B well-identified, DeepSeek V4 Pro weakly). The *compounding-error model* derives the cliff's threshold structure from per-round **critical-token-recall** against a per-family threshold θ_q, recovering its position only to first order: a mechanism, not a tight predictor. **Corollary 1 (Ceiling–Cliff Separation)** and **Corollary 2 (Information-Density Scaling)**, sharpened from H5 and H6, capture the no-shift and shape-scaling results, the latter via an AUC-based per-task θ_{info} (distinct from θ_q) validated on HotpotQA and MultiHop-RAG; an out-of-regime GPT-oss 120B diagnostic marks extended-reasoning planners as a structurally distinct case.

C3 (secondary): a catalogue of three retrieval-augmented-generation pipeline placements. Three placements (compress-first, retrieve-first, joint relevance-conditional routing) are compared under matched storage- and accuracy-bounded regimes. The pre-specified sign-flip is *not* observed; the narrowed finding is that compress-first beats retrieve-first on the single stack tested, not reproducing the LongLLMLingua [7] post-retrieval preference. As a single-stack non-reproduction orthogonal to the cliff and the bus, its full treatment is deferred to Appendix 7.5.

C4: A compression-aware memory bus, tuned by the cliff measurements. The principal engineering contribution is a memory bus that sits between the data sources and the planner, compresses every fragment in transit, and turns the token savings into lower inference cost: a FastAPI service with a tamper-evident SQLite audit log, a policy middleware enforcing a five-tier classification lattice over a 64-bit access-control mask, and a FAISS-CPU vector store, behind a single `Compressor` protocol. What makes it more than plumbing is that its compression layer is *driven by the findings*: a cliff-aware wrapper (**CAAC**) takes the per-family threshold θ_q and backs off, per fragment, to the most aggressive ratio that keeps critical-token-recall on the safe side of the cliff; and a summary-level inference-disclosure metric (H4, Section 4.10) caps how much a downstream reader can recover a protected fact from a compressed CONFIDENTIAL fragment, enforced at write time. The research content is the disclosure metric and the CAAC construction; the service operationalises them. Single-host throughput is benchmarked (Section 4.10.8); distributed scaling is follow-on work. The implementation is slated for deployment in the production workflows of the industry partner, TalentAdore, whose operating expense is dominated by token cost.

1.4.1. Reading Guide and Contribution Hierarchy

Table 1 is the one-page map of the whole thesis: it ties each research question to the artefact that answers it, the falsifiable hypotheses that operationalise it, and the one-line verdict, so that a reader can locate any result before reading the chapter that derives it.

The four contributions are not equally weighted: **H1** is the central, load-bearing result and the coordination cliff with its compounding-error model (C2) is the mechanism behind it; C1 is the enabling instrument; and the RAG catalogue (C3), the inference-disclosure metric and memory bus (C4), and CAAC are deliberately secondary, each operationalising the central finding in one deployment setting. A reader pressed for time should treat Chapter 4 Sections 4.4–4.6 and Section 4.7 as the core.

1.5. Thesis Outline

Chapter 2 surveys the prior work the four contributions build on: the transformer cost of context, the scaling era, the economics of inference, the training-free compression literature, multi-agent systems and agentic memory, retrieval-augmented generation and long-context benchmarks, and privacy in compressed retrieval, closing with the gap this thesis fills.

Chapter 3 presents the artefacts the evaluation runs on: the three-layer memory-bus architecture, the compressor framework, the C1 benchmark and its scoring pipeline, the critical-token-recall metric, and the compression cache.

Chapter 4 presents the empirical work: H1 (information preservation does not transferably predict coordination), H2 (a sharp cliff across 12 compressor-family cells), the compounding-error model that explains it, Corollary 1 (no cliff-position shift across planner scale and

Table 1: Contribution map. The four research questions, the artefact (C1–C4) that answers each, the falsifiable hypotheses that operationalise it, and the one-line verdict; all hypotheses are reported in Chapter 4. Throughout, *coordination* means multi-fragment solvability under compression (Section 1.2), *not* multi-round agent negotiation.

RQ / C	Plain-English question	Hyp.	Verdict (one line)
RQ1 C1–C2	Does keeping a single fragment answerable also keep several fragments *combinable*?	H1	**No**: single-fragment F1 does not predict coordination success (SUPPORTED).
RQ2 C2	Is there a sharp failure point as compression rises, and what fixes its position?	H2; Cor. 1; Cor. 2	A sharp cliff exists (SUPPORTED, 11/12 cells); its position is set by compressor+task, not planner scale (no shift detected); its shape scales with information density (SUPPORTED, n=3).
RQ3 C3	Where should compression sit relative to retrieval?	H3	Compress-first dominates on the tested stack; the sign-flip is *not* observed (*secondary*; Appendix 7.5).
RQ4 C4	Can the cliff findings drive a deployable memory bus that compresses safely and cheaply?	H4, CAAC	The bus compresses to the cliff-safe ratio (CAAC) and caps protected-fact disclosure (SUPPORTED, 3/4 compressors; destruction-driven).

architecture within a calibrated regime), H3 (RAG placement), H4 (inference disclosure), and Corollary 2 (the cliff shape scales with information density).

Chapter 5 synthesises the findings, presents CAAC as the constructive realisation of the model, and gives the limitations, significance, and future work.

All artefacts (manuscript source, code, configuration files, results CSVs, and the `docker-compose.yml` that brings the memory-bus service up locally) are in the reference repository at `https://github.com/Abdullah12has/Distributed-memory-bus-for-multi-agent-coordination`. Appendix 7 describes the reproducibility package and the one-command reproduction recipe.

2. BACKGROUND AND RELATED WORK

This chapter surveys the prior work the four contributions build on, in seven strands: the transformer architecture and the per-token cost compression targets (Section 2.1); the scaling-law trajectory (Section 2.2); the published energy and economic measurements of inference at scale (Section 2.3); the training-free compression literature (Section 2.4); multi-agent systems and agentic memory (Section 2.5); retrieval-augmented generation and long-context benchmarks (Section 2.6); and privacy-aware retrieval (Section 2.7). The consolidated gap statement is Section 2.8.

2.1. The Transformer Architecture and the Cost of Context

The large language models this thesis builds on are all transformers [16]. The transformer's defining mechanism, *self-attention*, computes N^2 pairwise compatibility scores for a sequence of length N, so prefill compute scales as $L \cdot (N^2 d + N d^2)$ over L blocks of hidden size d, and the key-value cache that decoding reads back grows linearly in N. The planners this thesis evaluates are decoder-only models (Qwen, Llama, DeepSeek, Phi-3), whose prefill cost $\approx 2PN$ (parameter count P, context length N) halves when the context halves: the lever context compression operates on.

Scaling laws [17, 18] gave the field a recipe to spend *training* compute productively but no corresponding recipe for *inference* compute, which is paid per query for the model's deployment lifetime and dominates training cost at production scale. Context compression is to inference cost what scaling laws are to training cost: a structural lever that converts a linear cost into a constant one.

2.2. The Scaling Era of LLMs

The trajectory from GPT-2 (1.5B, 2019) to GPT-3 (175B) to the 2025–2026 frontier (Llama-3, Qwen-2.5, DeepSeek V3/V4, Claude Sonnet 4.x, GPT-oss) marks four orders of magnitude of growth in deployed model size, each step raising per-query cost and the demand for context-engineering that controls how much input the model reads. The two-to-three-order gap between frontier-cloud and on-premises per-token cost is what makes compression an operational concern. Context windows grew in parallel (2K to multi-million tokens), but a model's effective use of them lags the nominal size (the "lost in the middle" effect [19, 20]), so tokens the model will not use well are pure cost. The model-scale trend is exactly what Corollary 1 of Chapter 4 tests: whether the cliff position depends on where on the scale curve the planner sits.

2.3. The Cost of Running LLMs at Scale

The motivation chapter frames the work around per-token cost; this section anchors that framing in published measurements.

Energy. Luccioni *et al.*'s *Power Hungry Processing* [1] measures inference wall-power across open-weight models (on the order of 0,1 Wh per inference for a 6–7B model, scaling near-linearly with size) and separates a per-request fixed cost from a per-token cost that dominates for long prompts. Samsi *et al.* [2] and Pope *et al.* [3] corroborate that energy and serving cost grow with the number of tokens processed once the prefill dominates: the long-prompt regime compression cuts.

Token economics. At ≈ USD 3 / 15 per million input / output tokens on a representative frontier tier (Anthropic Claude Sonnet; cheaper tiers such as GPT-4o-mini run ≈ 20× lower), one million 10,000-token calls per day is USD 30,000/day of input cost; halving the prompt saves half of that. On-premises marginal cost is two orders lower per token but the same lever applies, and the politeness tokens ("Hello", "Thanks", restated history) that opened Chapter 1 convert directly to kilowatt-hours regardless of whether they change the answer [5]. The TalentAdore production accounts confirm this pricing.

Multi-agent multiplier. Anthropic's multi-agent report [13] observes that token usage explains ≈ 80% of the performance variance they see, much of it shared context flowing between agents; the follow-on context-editing report [14] reports an 84% token reduction on a 100-turn web-search evaluation. These establish that the cost-of-context question is operationally pressing for production multi-agent deployments; what is missing, and what Chapter 4 supplies, is the structural answer to *how aggressively compression can be applied before coordination breaks*.

2.4. Context Compression: the Breakthrough Works

The training-free context-compression literature has matured in the past three years into four recognisable families, surveyed comprehensively by Li *et al.* [21]; the most aggressive recent variants push compression to the extreme of a single-token document representation (xRAG [22]). This subsection surveys the four families, with a comparison table at the end that maps the families against the design dimensions that matter for multi-fragment evaluation. Each family closes with a "what is missing in prior work" line that the rest of the thesis addresses.

2.4.1. Token-Level Pruning (the LLMLingua Family)

The first context-compression technique to be widely adopted was LLMLingua [6]: a small autoregressive language model scores tokens by their per-step perplexity contribution and drops those whose removal least disturbs the conditional distribution. LongLLMLingua [7] extends the coarse-to-fine ranking with a question-aware step to handle long retrieval-augmented contexts. LLMLingua-2 [8] trains an XLM-RoBERTa token classifier on data distilled from a more capable model, producing a task-agnostic compressor that is faster at inference than the original LLMLingua because it does not require running a generative model at compression time. The LLMLingua-2 classifier is the principal hard-prompt compressor of this thesis; its evaluation centres on NarrativeQA F1 within 2 pp and HotpotQA F1 within 3 pp of reported numbers [23, 24] as the calibration target.

What is missing. All three LLMLingua papers report single-agent QA-F1 retention; none measures whether the surviving content lets a planner combine information across multiple compressed fragments. H1 of Chapter 4 measures exactly that.

2.4.2. Selective and Instruction-Aware Compression

Li *et al.*'s *Selective Context* [25] prunes tokens with low self-information under the model's own conditional distribution; the technique requires no training but does require running a scoring forward-pass at compression time. Xu *et al.*'s *RECOMP* [26] compresses long retrieval contexts with a small task-aware extractor trained to summarise the retrieved passages for the downstream question. The instruction-aware filter of this thesis sits in the same family but is deliberately simpler: a TF-IDF pruning step, then a cross-encoder re-ranker (`BAAI/bge-reranker-base`), then a stop-word and short-token trim, with no LM in the loop. This gives a deterministic, training-free, low-cost baseline that the more sophisticated compressors must beat.

What is missing. This literature's task-awareness is question-conditioned, but in a multi-fragment workflow the fragment-level task is to preserve information a yet-unknown planner will combine with other fragments, the gap the instruction-aware filter and the multi-fragment evaluation address.

2.4.3. Learned Compression (Parameter-Cost Compressors)

Mu *et al.* [9] introduced *gist tokens*, special positions trained with a masked attention pattern so that the model learns to compress instruction prefixes into a small number of virtual tokens. Chevalier *et al.* [10] extended this with *AutoCompressor*, which fine-tunes a language model to process long documents segment by segment and produce summary vectors. Ge *et al.* [11] formalised the architecture as the *In-context Autoencoder (ICAE)*, which trains a LoRA-adapted encoder to produce a fixed number of memory tokens from an input context. Wang *et al.* [27] present the In-Context Former, a faster cross-attention variant. Rae *et al.* [28] introduced the Compressive Transformer architecture, which is the closest single-context analogue of the cliff bound this thesis derives.

What is missing. Learned compression requires training, introducing a training-budget asymmetry that muddies cross-compressor comparison. This thesis stays training-free so the cliff comparison turns on the algorithm rather than the training budget; re-evaluating learned compressors under the same methodology is named as follow-on work.

2.4.4. Extractive Small-LLM Span Selection

Recent work uses small instruction-tuned LLMs as extractive compressors that produce verbatim spans of the source rather than abstractive summaries. The candidate small LLMs are the Phi-3 family [29] and other $\sim$ 3B-parameter instruction-tuned models. Extractive compression is privacy-friendly because the compressed output is by construction a subset of the source tokens; this is one of the reasons it appears in this thesis as a compressor option.

What is missing. Published guidance on enforcing the extractive constraint reliably is thin; small LLMs insert function-word filler even when told not to. This thesis's Phi-3-Mini extractive compressor ships a verifier with a 15% novel-token tolerance and an LLMLingua-2 fallback (Section 3.3).

2.4.5. Comparison Table

Table 2: The compressor families this thesis surveys, along the design dimensions that matter for multi-fragment evaluation. The third column is the property H1 of Chapter 4 turns on: no prior compressor was evaluated on a controlled multi-fragment coordination sweep before this thesis.

Family	Training-free?	Task-aware?	Eval'd on multi-fragment?	Deployed in this thesis?
LLMLingua family [6–8]	yes (inference-only)	question-conditioned	no	LLMLingua-2
Selective Context, RECOMP	yes	yes	no	–
Gist Tokens / AutoCompressor / ICAE [9–11]	no (training required)	varies	no	–
Extractive small-LLM (Phi-3 family) [29]	yes (inference-only)	yes	no	Phi-3-Mini ext.
Instruction-aware filter	yes	yes	no	yes (this thesis)
Truncation (prefix-keep)	yes	no	no	yes (baseline)
This thesis (multi-fragment cliff sweep)	**yes**	**yes**	**yes**	**four compressors**

2.5. Multi-Agent LLM Systems and Agentic Memory

The multi-agent LLM-systems literature has grown rapidly since 2023. The most-cited frameworks are AutoGen [15] (an open-source framework for multi-agent conversation; originally appeared at the ICLR 2024 LLM-Agents Workshop, where it won Best Paper, and was subsequently published at the Conference on Language Modeling (COLM) 2024), MetaGPT [30] (ICLR 2024 main), CAMEL [31] (NeurIPS 2023), Reflexion [32] (NeurIPS 2023), MemGPT [33] (arXiv preprint, 2023), and Collaborative Memory [34] (Rezazadeh et al., 2025 preprint). Each framework implicitly answers how shared context flows between agents and what an individual agent needs to see. None of them formally bounds the information lost when that shared context is compressed.

The agentic-memory line of work [33, 35–39] addresses the related question of how to store and retrieve agent state across long horizons. The closest published academic analogue to this thesis's memory bus is MemIndex [38], which the three-layer memory-bus architecture of Chapter 3 explicitly extends. The MemGPT-style "LLM as operating system" framing [33] is related but is concerned with single-agent long-horizon memory rather than multi-agent shared state. Mem0 [36] is a production-oriented agentic-memory service.

What is missing. These frameworks treat compression as an implicit deployment optimisation; none measures the *multi-fragment* coordination-quality cost of compression structurally. This thesis fills that gap (the cliff sweep and the disclosure measurement); the orthogonal multi-round-agent gap is left to the AutoGen backend in `src/m6/orchestrator/` as future work.

2.6. Retrieval-Augmented Generation and Long-Context Benchmarks

Retrieval-augmented generation as a paradigm dates to Lewis *et al.* [40]. The landmark works since include RAPTOR [41] (recursive abstractive processing for tree-organised retrieval, ICLR 2024), GraphRAG [42] (NeurIPS 2024), HippoRAG [43, 44] (NeurIPS 2024 and follow-on), and Self-RAG [45] (ICLR 2024). The pipeline catalogue this thesis evaluates (P1, P2, P3) places compression on the retrieval critical path explicitly; the closest published analogue to the post-retrieval-compression pipeline (P2) is the LongLLMLingua configuration [7], and the closest published analogue to the joint relevance-conditional pipeline (P3) is the adaptive context-compression framework (ACC-RAG) of Guo *et al.* [46], which varies the compression rate by input complexity.

The long-context-benchmark literature ([19, 20, 47–50]) measures how effectively a model uses a long context. The "lost in the middle" result [19] is the canonical observation that information placed in the middle of a long context is recovered less reliably than information at the boundaries. RULER [20] extends the needle-in-a-haystack methodology to a broader set of long-context tasks. LongBench [47] is the standard multi-task benchmark. None of these benchmarks measures the multi-fragment-coordination task structure of this thesis; they measure single-agent long-context comprehension. They are complementary to the cliff measurement rather than competitive with it: long-context benchmarks measure the capacity of the model, while the cliff measurement measures the compression-quality effect on a task structure the long-context benchmarks do not include.

The canonical multi-hop question-answering benchmarks include HotpotQA [24] (EMNLP 2018), 2WikiMultiHopQA [51] (COLING 2020), and MultiHop-RAG [52] (COLM 2024). HotpotQA and MultiHop-RAG serve as the two real-data external arms on which Corollary 2 of Chapter 4 validates the information-density-scaling claim; the third data point in the $n=3$ comparison is the synthetic C1 family-a. The transfer is qualitative: the absolute cliff positions differ between synthetic and real benchmarks because the information density θ_{info} differs, but the cliff structure (existence, sharpness as a function of θ_{info}) transfers.

What is missing. RAG benchmarks measure retrieval quality and long-context benchmarks measure model capacity; neither captures the multi-fragment compression-quality effect the cliff measurement isolates. The H3 catalogue fills the compression-on-the-retrieval-critical-path gap.

2.7. Privacy in Agentic Memory and Compressed Retrieval

The privacy-aware-retrieval literature has two strands. The first protects the retrieval index itself, on the assumption that direct access to the index would leak protected facts. Examples include PrivacyRAG [53] and SecureRAG [54]. The second strand protects the model weights from training-data extraction, which is a property of the training pipeline rather than of the deployed retrieval-and-compression service.

The inference-disclosure metric this thesis contributes (C4) is distinct from both: it measures leakage *through the compressor itself*, on the assumption that the retrieval index is access-controlled but the compressed summaries are exposed to the planner and to any downstream reader the planner shares them with. The threat model is intuitive: an adversary who can query the planner can ask questions about the protected fact, and the disclosure rate is the fraction of those questions on which the planner's answer recovers the protected fact given that it has the compressed summary as evidence. The H4 measurement operationalises this in a controlled setup; the memory-bus policy layer of Chapter 3 operationalises the resulting disclosure budget as a deployment-time enforcement mechanism.

The CFedRAG framework [55] is a related piece of work on coordination of federated retrieval-augmented generation; it addresses the retrieval-index-level privacy question rather than the compressor-level disclosure question that this thesis takes up.

The closest published prior work that connects compression directly to privacy is SecurityLingua [56] (CoLM 2025), from the LLMLingua group, which uses prompt compression as a *defense* against jailbreak attacks: the compressor is trained to strip adversarial structure from inputs before they reach the model. SecurityLingua's framing is the dual of C4's: SecurityLingua applies compression as an input-side filter against the *adversary's* prompt, whereas C4 measures compression as an unintended leakage channel for protected facts in the *defender's* stored content, with a per-question disclosure rate against a fixed downstream reader. The two contributions are complementary rather than overlapping (a deployed memory bus could in principle combine SecurityLingua-style input filtering with C4-style summary-level disclosure auditing) but they measure different sides of the compression-versus-privacy interaction.

What is missing. No published work, to the author's knowledge, measures *summary-level* inference disclosure: the rate at which a reader recovers a protected fact from a compressed summary of content it has never seen. SecurityLingua [56] measures the dual (input-side defense); the privacy-aware-retrieval line protects the index, not the summary. This thesis fills that gap, and the policy layer converts the disclosure rate into a deployment budget.

2.8. The Gap This Thesis Closes

Four sentences, each one mapping to one of the four contributions of Chapter 1.

There is no shared multi-fragment coordination benchmark probing the cliff in a controlled, reproducible setup. C1 is the benchmark.

There is no published operational bound that predicts cliff position from compressor-side and task-side measurements within a defined regime. The compounding-error model and its cross-architecture frontier validation are the bound.

There is no honest catalogue of RAG-plus-compression pipeline placement under matched cost regimes, with the cost model explicit in EUR per workflow. The C3 catalogue is the catalogue, and its honest finding is that the assumed sign-flip does not appear.

There is no published academic measurement of inference disclosure through the compressor at the summary level. The H4 measurement, integrated with the memory bus's policy layer, is the measurement.

The thesis sits at the intersection of three threads: the long-context benchmark suite and the LLMLingua line of work (which measure orthogonal single-agent properties) and Anthropic's multi-agent report [13] (an industry observation, not a controlled measurement), and contributes the controlled cross-compressor multi-fragment evaluation none of them provides: it measures compression's effect on *multi-fragment* task success, where LLMLingua-2 [8] measures single-agent QA-F1 retention, LongLLMLingua [7] post-retrieval long-context QA, MemIndex [38] agent-memory storage, and the Anthropic report token-usage-versus-variance on one workflow.

3. SYSTEM DESIGN AND IMPLEMENTATION

This chapter presents the artefacts the empirical evaluation rests on: the reference memory bus, the compressor framework, the multi-fragment *C1* coordination benchmark, the coordination-success and critical-token-recall metrics that the cliff analysis turns on, the precomputed compression cache that decouples compression from evaluation, and the single-command reproducibility envelope. It gives a reader enough detail to reproduce the system from the reference repository without reading the code, and makes the design choices explicit so the results of Chapter 4 can be read against the constraints those choices imposed.

A scope reminder bounding every claim that follows: the planner is a deterministic regex information-extraction solver for the cliff sweep (H1, H2) and a single LLM call with all compressed fragments visible for the model-scaling sweep (Corollary 1, frontier, real-data transfer). The memory bus is *designed for* multi-agent integration via the AutoGen back-end in the source tree, but multi-agent operation is outside the empirical scope (Section 3.8).

3.1. The Memory-Bus Reference Implementation

3.1.1. Three-Layer Architecture

The reference implementation organises the memory bus into three layers that match the access pattern of a fragment-passing workflow:

1. An **access layer**, exposing `read`, `write`, `subscribe`, and `audit` endpoints via FastAPI and enforcing per-request access-control checks through a policy middleware.
2. A **compression layer**, providing a pluggable `Compressor` protocol behind which six concrete variants live: an *identity* no-compression control, the *LLMLingua-2* [8] token classifier, an *instruction-aware filter* based on TF–IDF pruning plus a cross-encoder re-ranker, a *Phi-3-Mini extractive* span selector implemented over a local Ollama endpoint, a *truncation* prefix-keep baseline, and *CAAC*, a cliff-aware adaptive wrapper that backs off compression when per-fragment recall would violate the compounding-error model's safety threshold (Section 3.3.5, full discussion in Chapter 5).
3. A **storage layer**, providing three protocols (`AuditLog`, `Scratchpad`, and `VectorStore`) implemented for the evaluation by SQLite in WAL mode with an append-only audit table, an in-memory time-to-live dictionary, and FAISS-CPU with an HNSW index.

Figure 1 summarises the three layers. Each component dependency in the source tree is one-directional: the compression layer does not import from the access layer, and the storage layer does not import from either. This layering means the storage backends can be replaced without touching the compression or access layers.

The three-layer structure is a direct extension of MemIndex [38], the FCG group's baseline architecture for distributed agent memory. The two material differences are that compression is promoted to a first-class layer rather than treated as a downstream optimisation, and that per-slot tags are part of the data model rather than out-of-band metadata.

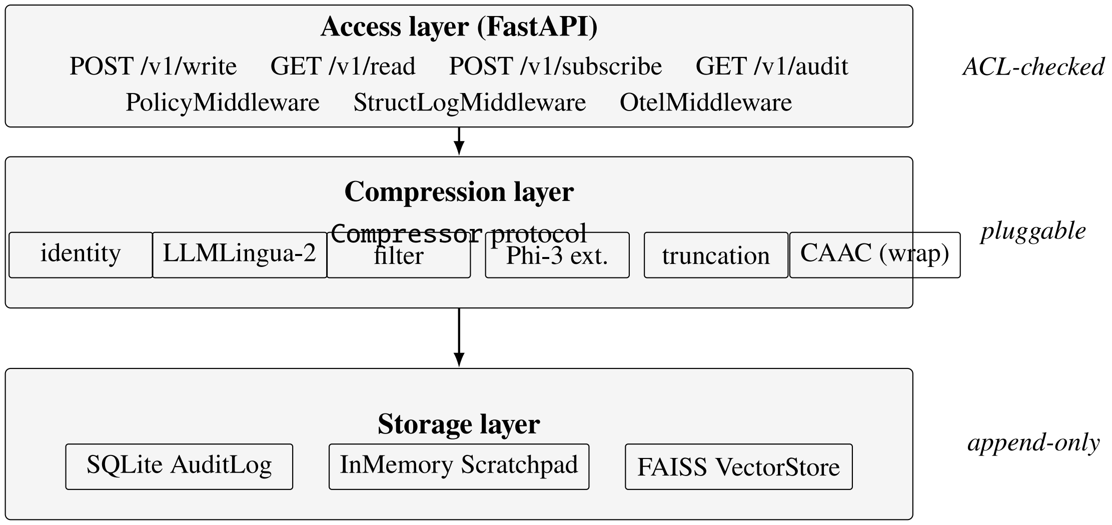

Figure 1: Three-layer architecture of the memory-bus reference implementation. Each layer depends only on the layer below it, and each storage component is hidden behind a `Protocol` so that the storage backends can be replaced without touching the access or compression layers. CAAC is drawn inside the compression layer for convenience but is a wrapper: it composes any of the other compressors with a recall-side back-off rule (Section 3.3.5).

3.1.2. Data Flow for a Write

A `POST /v1/write` carries the fragment text, its tag vector, and an optional task hint. The policy middleware parses the principal from the `X-M6-Principal` header and checks that its access-control mask is a superset of the fragment tag's and its classification at least the fragment's. On success the service compresses the fragment to a `CompressedSlot`, stores it in the scratchpad, adds any embedding to the FAISS index, and appends a `WRITE/OK` audit row whose `chain_hash` = SHA-256(`prev_hash` ‖ `payload_hash`); the response returns the slot id, hex chain hash, and audit row id. A failed check appends a `DENY` row (slot `deny-<uuid8>`) with a 60-second per-(subject, fragment-id) deduplication so a client hammering denied writes cannot inflate the log.

3.1.3. Data Flow for a Read

A `GET /v1/read/{slot_id}` flows symmetrically: the middleware attaches the principal, the service looks up the slot, checks the principal–tag relationship, and either returns the slot with its audit row id or raises a policy-denied / slot-not-found error. Read misses are audited as `READ/ERROR` rows with the same 60-second per-(subject, slot-id) deduplication. Appendix 7 gives an end-to-end `curl` trace.

3.1.4. Tags, Classifications, and the Audit Log

Every fragment and every compressed slot carries a tag vector recording its provenance and access-control state. The tag vector has four fields: a `uint64` access-control mask, a five-tier classification level (PUBLIC < INTERNAL < CONFIDENTIAL < RESTRICTED < SECRET) drawn from enterprise data-classification practice (e.g., ISO/IEC 27001 organisational schemes); the broader risk-management framing follows the NIST AI Risk Management Framework [57], but the five-tier lattice itself is enterprise convention and not prescribed by NIST AI RMF (which defines Govern/Map/Measure/Manage functions rather than a labelled classification lattice). The remaining two fields are a tuple of source identifiers recording provenance and a tuple of parent-slot identifiers recording which previously-compressed slots this slot inherits from. When slots are merged, the union's access-control mask is the bitwise OR of the parents' masks and the union's classification is the maximum; the policy predicate is symmetric for reads and writes.

The audit log is an append-only SQLite table; each row records the `event_type` (WRITE/READ/SUBSCRIBE/DENY/COMPRESS/EVICT), `slot_id`, requester ACL mask and classification, result, and a tamper-evident hash chain (`chain_hash` = SHA-256(`prev_hash` ‖ `payload_hash`)). An `INSTEAD OF UPDATE` and an `INSTEAD OF DELETE` trigger refuse any mutation of an existing row, and a unit test drops the trigger, edits a `payload_hash` at the byte level, and confirms that the `verify()` chain-walker reports failure.

A second SQLite table records every paid model call (provider, model, token counts, EUR cost, wallclock), sharing the audit database for transactional consistency. Pricing follows a representative frontier rate (Anthropic Claude Sonnet tier, USD 3/15 per million input/output tokens, ≈ EUR 2,76/13,80 at 0,92 USD/EUR) against an on-premises marginal ≈ EUR 0,05 per million tokens [58]; the ledger feeds the EUR-per-workflow cost model of the RAG catalogue (Section 3.2).

3.2. Retrieval-Augmented Generation Pipeline Catalogue

The H3 analysis (Appendix 7.5) compares three pipeline placements over the shared FAISS-CPU + HNSW + `bge-large-en-v1.5` stack: **P1** compress-first (compress the corpus, then index it), **P2** retrieve-first (index the full corpus, then compress the top-k, the classical LongLLMLingua [7] configuration), and **P3** a joint relevance router (pass high-scoring fragments verbatim, compress mid-scoring, discard low-scoring; after ACC-RAG [46]). Because the pre-specified sign-flip did not appear and the cost model does not meter compression compute, H3 is reported as a secondary result in Appendix 7.5.

3.3. Compressor Framework

Every compressor variant conforms to a single Python `Protocol` with four methods: `compress(fragment, target_ratio)` returning a `CompressedSlot`; `decompress(slot)` returning a best-effort text reconstruction; `embed(slot)` returning an optional embedding for vector-store indexing; and `model_card()` returning a structured card containing the compressor identifier, family, base model, default target ratio, and provenance hashes. The thesis ships six concrete implementations. The identity compressor is the no-compression control that every experiment needs to guard against runtime-bug confounds.

The remaining five compressors fall into four families: a token-level hard-prompt filter (LLMLingua-2); an instruction-aware heuristic filter; an extractive small-LLM span selector (Phi-3-Mini extractive); a trivial prefix-truncation baseline; and a cliff-aware adaptive wrapper (CAAC). *All five are training-free*, which is a deliberate design choice: the thesis isolates the compression mechanism from any training-data effect and lets the cross-compressor comparison turn on the algorithm rather than on training-budget asymmetry.

3.3.1. LLMLingua-2 (Token-Level Classifier)

The LLMLingua-2 wrapper is a thin adapter around the `llmlingua` package [8]. Construction takes a target compression ratio; the package is asked to retain $1/r$ of the tokens, scored by a small XLM-RoBERTa classifier distilled from a more capable teacher. The force-tokens list (sentence boundary marks) is validated against the wrapped tokenizer's vocabulary at initialisation and unknown tokens are dropped with an explicit log line. PyTorch runs the wrapped XLM-RoBERTa model on both CPU and Apple's Metal Performance Shaders backend, so the compressor runs natively on Apple Silicon without modification.

The empirical token-recall curve $q(r)$ produced by LLMLingua-2 under critical-token-recall measurement (Section 3.4.5) is smooth and monotonic across the sweep ratios used in Chapter 4, which is the property the compounding-error model presumes.

3.3.2. Instruction-Aware Filter

The instruction-aware filter is a pure-Python heuristic baseline. It operates in two stages. The first is a sentence-level filter that ranks sentences by their relevance to the task hint using a cross-encoder reranker (`BAAI/bge-reranker-base`) when available and a lexical-overlap fallback when not. The second is a token-level trim that drops English stop-words and short tokens ($\leq$ 2 characters) until the target compression ratio is reached. The stop-word list is encoded explicitly so that the trim is deterministic given the task hint, the fragment, and the target ratio: a subtle bug showed that dropping stop-words in Python set-iteration order introduced run-to-run variance.

The filter's reranker preserves task-keyword tokens that simpler token-level methods drop, which is why its $q(r)$ curve on the multi-step retrieval family (Section 3.4, family-c) decays gracefully rather than collapsing: chain-reference tokens like "entry" and "FINAL" survive aggressive compression.

3.3.3. Phi-3-Mini Extractive Span Selector

The Phi-3-Mini extractive compressor is the only thesis compressor that uses an LLM at inference time. It runs the Phi-3-Mini-3.8B instruction-tuned model [29] via a local Ollama endpoint [59] with a structured prompt that asks for verbatim contiguous spans from the source. The post-hoc verifier `Phi3ExtractiveCompressor._verify_extractive` checks the output: every token in the output must appear in the source. Two loosenings of the strict verifier were necessary in practice. First, a 15% novel-token tolerance: small language models frequently insert function words ("the", "is", "and") even when explicitly asked not to. Second, a post-hoc `_strip_novel_tokens` pass that removes any token not in the source after the verifier passes; this guarantees zero novel tokens in the final output even if the model produced some during generation. Outputs that fail even after stripping fall back to LLMLingua-2 at the same target ratio.

Phi-3-Mini extractive has a *compression ceiling*: span-level selection, the stripping pass, and the LLMLingua-2 fallback together bound the achievable ratio at about 2,5×, regardless of the requested target. This is documented as a limitation in Chapter 5 and is the reason all evaluation reports *achieved* compression ratio (the ratio actually delivered) alongside the requested target ratio.

3.3.4. Truncation (Prefix-Keep Baseline)

The truncation compressor keeps the first $1/r$ fraction of tokens by whitespace split. It is deterministic by construction and achieves the requested ratio exactly. Its purpose is the lower-bound argument: every more sophisticated compressor must beat truncation on the coordination metric to justify its cost.

Truncation has a peculiar property: its single-agent question-answering F1 is competitive with the more sophisticated compressors at low to moderate compression ratios because the prefix tokens are naturally representative. Its coordination success collapses earlier than the others, because the tokens removed at the tail include the task-critical content that the planner needs to combine. This dissociation is part of the H1 finding: a token-overlap quality metric over-reports the utility of even the most naive compressor.

3.3.5. Cliff-Aware Adaptive Compression (CAAC)

CAAC is a wrapper, not a fifth compressor family. Given an inner compressor (LLMLingua-2 in the headline experiments; any of the others in the ablation) and a per-family recall threshold θ_q, CAAC runs the inner compressor at the requested target ratio, measures the resulting critical-token-recall, and if recall falls below $\theta_q^{1/N}$ binary-searches downward for the largest ratio that keeps recall above the threshold. A `min_ratio` floor of 1,5× prevents CAAC from abdicating to no-compression on hard fragments. The full algorithmic and empirical discussion lives in Chapter 5 together with the constructive framing that CAAC is the operational realisation of the compounding-error bound. Here it suffices to note that CAAC consumes the same `Compressor` protocol as the other variants, which is what makes its drop-in evaluation in Chapter 4 straightforward.

3.4. The C1 Coordination Benchmark

3.4.1. Why a Synthetic Benchmark

The coordination cliff τ^* is a position on a curve, and detecting it requires sweeping ratios densely from 1× to 16× with everything else held constant. Real benchmarks (HotpotQA, MultiHop-RAG) confound the cliff with task difficulty, retrieval distribution, and per-instance variance, so the thesis characterises the cliff on a synthetic generator first (where information density is placed by construction) and validates the transfer on real benchmarks (Corollary 2). The C1 generator produces 150 instances across three families from a single seed, with the config hash recorded in the manifest so any output CSV traces back to its generator inputs.

3.4.2. Three Workload Families

Family-a: cross-document fact aggregation. Eight source documents, one per Vignette 3.7 institutional system [12], each carrying a recorded-hours and an approved-budget number from a seed-fixed distribution; the planner aggregates both across the eight, scored by a regex extraction pipeline against the deterministic arithmetic sum. Family-a is the densest ($\theta_{\text{info}} \approx 0{,}97$): every multi-digit number is task-critical and the cliff is sharp, so it is where the compounding-error model is most directly testable.

Family-b: constraint-satisfaction planning. N sub-tasks with per-task loads and K capacity-bounded workers; the planner must assign every sub-task within capacity. The generator guarantees at least one feasible assignment (capacity-respecting greedy), and the scorer is a *feasibility checker*: any valid assignment scores 1. Constraint tracking is hard for ≤ 8B planners even without compression, so the family-b baseline p_0 is low for small planners and the cliff is undetectable there: a *floor effect* Chapter 4 treats explicitly.

Family-c: multi-step retrieval. A linear chain of fragments, each carrying a pointer to the next (*entry X*) or, at the leaf, the answer (*FINAL-XXXX*); chain lengths 2–4, padded with 4–8 distractor sentences, scored against the leaf answer. Family-c is the most distributed of the three: many tokens are distractors, information density $\theta_{\text{info}} \approx 0{,}62$ (distinct from the recall threshold θ_q; see Section 4.11), and the cliff is gradual; it is the family on which the instruction-aware filter's reranker, which preserves chain-reference keywords, can avoid a detectable cliff entirely.

Each family contributes 50 instances. The three families together span a deliberate range of information densities so that the cliff mechanism (a recall threshold θ_q) and the cliff-position shift mechanism (information density θ_{info}) can be separated; this separation is what Chapter 4 formalises as Corollary 2.

3.4.3. Tag Distributions for the Privacy Axis

The C1 generator supports three synthetic tag distributions (*uniform*, *skewed*, *hierarchical*) used by H4 to probe the governance-stringency dimension. The hierarchical distribution sets higher classification levels to imply strict supersets of the lower-level bits, matching enterprise tag hierarchies and letting H4 disentangle classification level from access-control mask.

3.4.4. Coordination Success and Supporting Metrics

The coordination scorer is a pure function of the workload and the planner's output; it does not call any LLM. It returns one primary metric (a per-instance binary *coordination success*) and four supporting metrics: the F1 over extracted answer tokens versus ground truth, the achieved compression ratio ($|\text{source}|/|\text{compressed}|$), generic token-recall (the fraction of source tokens preserved in the compressed text), and *critical-token-recall* (CTR; Section 3.4.5). Aggregation across workloads × seeds is by arithmetic mean; 95% confidence intervals are computed by workload-level bootstrap, never by per-row bootstrap which double-counts the seed dimension.

The reason the scoring pipeline never calls an LLM is the same reason H1 and H2 use a deterministic regex parser as the planner: isolating the compression-quality signal from LLM run-to-run variance. The trade-off is that the cliff measured here is an upper bound on what an LLM-based scorer would measure; the trade is in favour of clean attribution rather than ecological validity, and Chapter 5 discusses the limitation.

3.4.5. *The Critical-Token-Recall Metric*

A standard token-recall measurement $q_{\text{token}}(r) = |T_{\text{compressed}} \cap T_{\text{source}}|/|T_{\text{source}}|$ counts every preserved token equally. This proved a poor proxy for task-relevant information preservation: the Phi-3-Mini extractive compressor at low ratios preserves common function words like "the" and "and" but drops some critical numeric tokens. Its token-recall is high; its coordination success is low.

The critical-token-recall (CTR) metric is the family-specific restriction:

$$q_{\text{CTR}}(r) = \frac{|T^{\text{crit}}_{\text{compressed}} \cap T^{\text{crit}}_{\text{source}}|}{|T^{\text{crit}}_{\text{source}}|},$$

where T^{crit} is the set of *task-critical* tokens for the family:

- Family-a: multi-digit numeric tokens (at least two digits, to exclude single-digit noise);
- Family-b: all numeric tokens (load and capacity values);
- Family-c: chain-reference tokens (*entry-X* patterns) and the literal *FINAL* marker.

CTR is what the compounding-error model uses as the per-round retention rate $q(r)$; CAAC uses CTR as its back-off signal; and the predicted-versus-empirical τ^* figure in Chapter 4 uses CTR to derive θ_q. Generic token-recall is reported for backwards compatibility with prior analyses but is not used in the verdict pipeline.

3.5. Compression Cache

Compression is the slowest evaluation step (Phi-3-Mini extractive is a $\sim$ 11 second Ollama call per fragment), and the cliff sweep touches $\approx$ 30,000 (compressor, ratio, workload, task-hint) cells, so a precomputed cache (`CompressionCache / CachedCompressor` in `src/m6/compressors/cache.py`, keyed on (compressor, r, fragment_id, hash(task_hint))) is populated once on the GPU and consumed as JSON thereafter. This decouples compression from evaluation, letting H1, H2, the frontier sweep, and the CAAC ablation run on a laptop, and guarantees every run sees identical compressed text for a cell, so run-to-run differences are protocol, not compressor stochasticity (cache hit rate 100% by construction).

3.6. Inference Backends

The unified `InferenceBackend` protocol exposes a single `complete(prompt, max_tokens, temperature, stop)` method together with an optional `embed(text)` method. Three concrete backends are used in the evaluation. The first is a local Ollama endpoint [59] that hosts Qwen-2.5-Instruct at the three local sizes (1.5B, 3.8B, 8B) used in the model-scaling sweep, Phi-3-Mini-3.8B for the extractive compressor, and Llama-3.1-8B-Instruct [60] as the local protected-fact reader. The second is the OpenAI-compatible Featherless API used for the frontier validation (Qwen-2.5-72B-Instruct, DeepSeek V4 Pro). The third is the OpenAI API for the GPT-class frontier arm (excluded from the calibrated regime per Chapter 5, retained on disk as a noncanonical diagnostic). All API calls are routed through the cost ledger so that the EUR-per-workflow figures in Chapter 4 are auditable.

3.7. Design Choices and Alternatives Considered

The artefacts above are not a default stack; each tool and parameter was a choice with a credible alternative, catalogued here so the results of Chapter 4 can be read against the decisions that made them. The per-hypothesis falsifiability bars are discussed at the point of pre-specification (Chapter 4 Section 4.1).

3.7.1. Tooling: Storage, Retrieval, and Serving Stack

The serving stack is chosen for single-host reproducibility: **FAISS-CPU** [61] with an HNSW index (over Qdrant/Chroma/pgvector) so the reference service ships as one Docker image with no external database and no GPU dependency for retrieval; `BAAI/bge-large-en-v1.5` embeddings with a `bge-reranker-base` re-ranker (open-weight, digests in `docs/MODEL_CARDS.md`); **FastAPI**/Starlette for the access layer; **SQLite-WAL** for the audit log (one-file footprint, concurrent reader–writer, a replicated multi-host chain named as future work); **Ollama** for local LLM serving (uniform model lifecycle across the Apple-Metal and CUDA hosts behind one `complete()` call, Section 3.6); and the OpenAI-compatible **Featherless** endpoint for the frontier arm, chosen for its flat per-token price so the EUR cost model compiles without per-provider markup tracking.

3.7.2. Compressors Considered but Not Swept

Three compressor families from the survey were considered and excluded, recorded so the omissions are deliberate. **Selective Context** [25] requires a scoring forward pass at compression time (asymmetric compute with the swept four) and its design point (task-aware lexical pruning with no end-to-end LM) is already covered by the instruction-aware filter. **RECOMP** [26] and **learned compression** (gist tokens [9], AutoCompressor [10], ICAE [11]) require fine-tuning, which would confound the deliberately training-free cliff comparison with training-budget asymmetry (re-evaluating them is named as follow-on work). **xRAG** [22] represents a document as a single token ($> 100\times$), beyond the 1–16$\times$ regime where the cliff sits. For the extractive compressor's base model, Phi-3-Mini was chosen over Llama-3.2-3B and Mistral-7B on pilot extractive-verifier pass rates ($\sim$ 75% vs $\sim$ 62%/$\sim$ 71% pre-fallback) at the smallest deployment footprint.

3.7.3. Threshold and Sweep-Grid Parameters

The cliff measurement turns on a few numerical parameters. The **ratio grid** $\{1, 2, 3, 4, 5, 6, 8, 10, 12, 16\}\times$ is denser at the low end where the dense-family cliff sits; a fine-grid re-resolution (Chapter 4 Section 4.5) later showed the $r \in \{1, 2\}$ gap is itself too coarse for the deterministic-solver family-a cliff near $r = 1{,}1$, with $\mathbf{r}_{\max} = \mathbf{16}$ because beyond it every compressor's CTR is near zero on family-a. 50 **instances per family,** 5 **seeds per cell** (250 pairs per cell) is the smallest sample at which the workload-level bootstrap CI clears the H1/H2 bands at the budgeted compute; there is no formal a-priori power analysis, named as a limitation in Chapter 5. θ_q **at** $0{,}5 \times p_0$ reads off "coordination has collapsed" without a separate calibration sweep and sits below the H2 30%-drop bar; the CAAC ablation finds the coordination plateau invariant to $\theta_q \in \{0{,}6, 0{,}7, 0{,}8\}$. The **bootstrap** uses BCa intervals [62] at $n_{\text{boot}} = 10{,}000$ on the verdict pipelines and 500 on the θ_q and frontier-τ^* pipelines, whose per-iteration curve fit is the budget driver.

3.8. Scope of Evaluation: the Multi-Fragment Proxy

The empirical protocol uses two planners, isolating the compression-quality signal from LLM variance: a deterministic regex information-extraction solver for H1/H2, and a single LLM call with all compressed fragments visible for Corollary 1, the frontier validation, and the real-data transfer (demonstrating that the cliff structure the solver measures appears identically under a non-trivial language model). The AutoGen v0.4 multi-round backend [15] in `src/m6/orchestrator/` integrates with the bus but is not used: multi-round negotiation variance dominates the compression signal at the swept ratios, so the backend demonstrates that the bus is multi-agent-ready rather than serving as the measured planner. Coordination success (Section 3.4) is therefore the structural property that the planner must combine information across fragments, not multi-round communication quality, which Chapter 5 names as future work.

3.9. Compute Envelope and Reproducibility

The empirical work runs on an Apple M4 Pro 48 GB workstation (local Ollama planners, deterministic-solver pipeline, and every figure when the compression cache is in place) and an RTX 5090 32 GB host (compression precompute, the 8B model-scaling sweep, the frontier API sweep, and the HotpotQA / MultiHop-RAG transfer). Total wallclock for the canonical pipeline is $\sim$ 30 GPU-hours plus $\sim$ 12 hours of laptop time. The reproducibility package is the complete repository: a `docker-compose.yml` for the memory-bus service, a release tag, model and data cards, and one-command `make` targets for every headline figure; one CSV per hypothesis run plus a `manifest.yaml` recording the resolved configuration, the config hash, the git revision, and the platform. Appendix 7 lists the commands and the full compute stack.

4. EXPERIMENT DESIGN AND PROTOCOL

This chapter presents the empirical work that this thesis turns on: the cliff sweep on which H1, H2, and the compounding-error model are based; the model-scaling sweep whose result is reported as Corollary 1; the frontier validation that promotes the model-independence claim from small-model evidence to a cross-architecture result inside a defined calibrated regime; the retrieval-augmented-generation pipeline catalogue (H3); the summary-level inference-disclosure measurement (H4); and the information-density scaling result on real benchmarks reported as Corollary 2. The chapter is organised around the falsifiable claims rather than the chronology of the runs; canonical results directories under `results/` are referenced for each section so that every number is traceable to an immutable artefact on disk.

4.1. Pre-Specification and Refinement

The six investigations of this chapter were pre-specified with falsifiable criteria in `configs/experiments/h{1-6}.yaml` and `plan-v3.md`. The six hypotheses and the statistical protocol were committed to the project repository on 2026-05-21, prior to the data-generation runs that produced the headline numbers reported in this chapter (the final sweeps ran from 2026-05-26 onward). This is a pre-specified protocol committed to version control, not a third-party-frozen pre-registration (e.g. OSF or AsPredicted). The criteria for H3, H5, and H6 were refined after the sweeps; the refinements are documented as ADR-006–008 in the project repository. The three-tier structure used in the rest of the chapter is as follows.

Three hold as pre-specified. H1 (workload-level source-retention F1 does not transferably predict coordination success), H2 (a sharp cliff at $\geq$30% drop with $p < 0{,}05$ on $\geq 7/9$ cells), and H4 (compression reduces inference disclosure with baseline above priors).

Two sharpened. H5's monotonicity prediction resolved into the invariance result **Corollary 1 (Ceiling–Cliff Separation)**, Section 4.7: τ^* is determined by the compressor and the task within a calibrated regime, while planner scale moves the ceiling p_0. H6's transfer prediction resolved into **Corollary 2 (Information-Density Scaling)**, Section 4.11: the cliff position varies with task information density θ_{info}.

One narrowed. H3's regime sign-flip predicate ($\geq$5 pp F1 effect with opposite sign across storage- and accuracy-bounded regimes) is not satisfied; compress-first dominates on the tested stack in both regimes (Section 4.9).

4.1.1. Falsifiability Criteria as Practitioner-Relevant Minima

Each numerical bar was chosen as the smallest effect a practitioner relying on this thesis would care about, not an arbitrary statistical threshold. **H1** ($\rho < 0{,}6$, workload-level): Cohen's "less-than-moderate" rank correlation, above which an F1-based compressor ranking would still be operationally useful. **H2** ($\geq$ 30% relative drop, Holm-corrected $p < 0{,}05$): a 10% drop sits within run-to-run variance and a 50% drop is obvious without measurement, so 30% is the deployable/undeployable boundary; Holm is applied within the 12-cell (compressor, family) family. **H3** ($\geq$ 5 pp F1 with opposite sign across regimes): the smallest F1 gap the EUR cost model makes meaningful, with opposite sign the only thing that would justify regime-conditional deployment. **H4** (paired-bootstrap $p < 0{,}05$, positive direction): paired because the same workloads appear in all conditions; positive is the privacy direction. **Corollary 1** (±20% TOST band): the strict cliff-detectability tolerance, reported alongside the ±50% permissive default in `validate_model_independence`. **Corollary 2** ($\Delta\theta_{\text{info}} \geq 0{,}1$): the smallest density gap producing a qualitatively distinct cliff shape; the empirical gaps (0,48, 0,59) clear it several-fold. Tool and grid choices live in Chapter 3 Section 3.7, metric choices in Section 4.3.

4.2. Experimental Protocol

4.2.1. The Cliff Sweep

The cliff sweep underlies H1, H2, and the calibration step of the compounding-error model. Four compressors (LLMLingua-2, Phi-3-Mini extractive, the instruction-aware filter, and a truncation baseline) are evaluated at ten compression ratios $\{1, 2, 3, 4, 5, 6, 8, 10, 12, 16\}\times$ on the 150 C1 workloads across the three families. Five seeds per cell give a total of approximately 30,000 evaluation cells per run. The canonical results directory is `results/h1_h2_v2/`; the earlier three-compressor variant `results/h1_h2_final/` is retained on disk for backwards-compatibility checks but is not the source of the headline numbers below. The truncation baseline is the lower-bound argument: any compressor that does not beat truncation on the coordination metric is paying training or inference cost for no coordination-quality return.

Every cell records the coordination-success binary, the source-retention F1 (token-overlap F1 between the compressed fragments and the original source text: the `qa_f1` column H1 analyses, *not* answer-level QA-F1 against a gold answer), the achieved compression ratio, the generic token-recall, and the critical-token-recall described in Section 3.4.5 of Chapter 3. The compression cache (Section 3.5) guarantees that every cell sees the exact same compressed text for a given (compressor, ratio, workload) tuple, so within-cell variance is exclusively planner-side variance, not compressor stochasticity.

4.2.2. Statistical Protocol

The statistical protocol is fixed in advance. Effect sizes are computed at the workload level, never the cell level: per-workload means first, then 95% BCa bootstrap CIs by resampling workloads [62] ($n_{\text{boot}} = 10{,}000$ on the H1/H2/H3/H4 verdict pipelines, 500 on the θ_q and frontier-τ^* pipelines whose per-iteration curve fit is costlier). The two-cell drop test is the paired Wilcoxon signed-rank test [63] (not Mann–Whitney [64], whose independence assumption the shared workloads violate), and multiple comparisons use Holm's procedure [65] *within each hypothesis's family* (12 cells for H2, the condition pairs for H4, the pipelines per regime for H3); joint correction across the independent hypotheses would be over-conservative.

The H1 verdict statistic is the *workload-level* Spearman correlation ($n = 150$, one mean Δ-pair per workload); a pooled correlation over $n = 150 \times 9$ (workload, ratio) measurements would over-state significance by an order of magnitude (the nine ratios are repeated measures), so the pooled column in Table 3 is non-inferential. The verdict rests on the BCa bootstrap CI over workloads excluding 0,6.

Cliff position τ^* is extracted by fitting both a piecewise-linear model and a logistic

$$f(r) = p_0 + \frac{p_\infty - p_0}{1 + e^{-k(r-\tau^*)}}$$

to each cell's coordination-success-versus-ratio curve and selecting the model with the lower root-mean-squared error. The piecewise fit is constrained to an interior 10% margin from the sweep boundaries. This prevents the optimiser from parking τ^* at $r_{\max} = 16$, a degenerate fit the unconstrained optimiser would otherwise select. 95% bootstrap CIs on τ^* are computed by resampling the underlying workload means before the curve fit.

4.3. Choice of Evaluation Metrics

Every metric in this chapter is a choice; a more conventional one would in some cases have produced a different verdict. The verdict-critical metrics, binary coordination success, source-retention F1 (`qa_f1`), critical-token-recall, workload-level Spearman ρ, and BCa-bootstrap / paired-Wilcoxon / Holm statistics, are each defined and motivated where they enter the pipeline; this section consolidates the workload-level versus cell-level statistical-unit decision the H1 verdict turns on. Falsifiability bars are in Section 4.1; tool and grid choices in Chapter 3 Section 3.7.

4.3.1. Exact-Match Coordination versus Similarity Metrics

The verdict metric is a per-instance binary *coordination success* against a deterministic ground truth, not a semantic-similarity score (BERTScore [66], ROUGE [67]) nor an LLM-judge pipeline (RAGAS [68], DeepEval [69]). Three reasons. First, every C1 task has a verifiable ground truth (an arithmetic sum, a capacity-feasible assignment, a chain-leaf answer), so a graded similarity score would add noise without resolving correctness: the planner either recovers the answer or it does not. Second, byte-level reproducibility (Section 4.12, Tier 1) requires a deterministic scorer; an LLM-judge would reintroduce exactly the planner-side stochasticity the deterministic solver is designed to isolate (Chapter 3 Section 3.8). Third, source-retention F1 is retained *because* it is the metric the prompt-compression literature optimises; that is precisely what H1 needs in order to show the two disagree, and substituting a similarity metric would blunt the comparison rather than sharpen it. Semantic metrics remain the right tool for the open-ended generation tasks (summarisation, dialogue) the compression literature targets; they are mismatched to a coordination task with a checkable answer.

4.3.2. Workload-Level versus Cell-Level Statistics

A single methodological decision sits behind every effect-size figure in this chapter: the inference-valid statistical unit is the *workload*, not the (workload, seed) cell or the (workload, ratio) pair. The reason is mechanical: the five seeds on each (compressor, ratio, workload) cell are repeated measures of the same underlying coordination event, and the nine ratios per workload are repeated measures of the same underlying compressor behaviour. Treating either as independent draws, which a pooled correlation or a pooled bootstrap would do, inflates the effective n by roughly an order of magnitude and over-states statistical significance.

The H1 verdict in particular illustrates this: as Section 4.4 reports, two of the four compressors (LLMLingua-2 and truncation) drop from a pooled $\rho \approx +0{,}38$ at $n = 1350$ to a workload-level $\rho \approx 0{,}03$ at $n = 150$. The verdict (all CIs exclude 0,6) holds in both views, but only the workload-level magnitudes correspond to a relationship a practitioner could exploit. The pooled column is reported as a non-inferential comparison only, and the H1 verdict table carries this distinction in its caption.

4.4. H1: Single-Fragment Information Preservation Does Not Transferably Predict Coordination Success

H1. Single-fragment information preservation under compression is not a transferable predictor of multi-fragment coordination success: workload-level Spearman $\rho(\Delta_{\mathrm{qa}}, \Delta_{\mathrm{coord}}) < 0{,}6$ for every compressor, with 95% bootstrap CIs excluding 0,6 from above. **Verdict: SUPPORTED.** At the workload level (the inference-valid unit) the instruction-aware filter is strongly negative ($\rho = -0{,}82$), Phi-3-Mini extractive moderately positive (+0,32), and LLMLingua-2 and truncation near zero (+0,03, +0,05), even though the latter three look more correlated ($|\rho| \approx 0{,}38$–$0{,}59$) when ratios are pooled. The honest statement is the negative one (*source-retention F1 change does not positively predict coordination change for any compressor*) plus a warning that pooled single-number correlations are distorted by pseudo-replication. The per-family decomposition (Table 4) confirms there are zero significant *positive* within-family correlations.

4.4.1. Result

The H1 single-fragment metric is *source-retention F1*: token-overlap F1 between the compressed fragments and the original source text (the `qa_f1` column), *not* answer-level QA-F1 against a gold answer; it is the proxy for the single-agent quality metric the prompt-compression literature optimises [7, 8]. Table 3 reports the Spearman correlation between its change and the change in coordination success, at the workload level (inference-valid) and pooled-over-ratios (comparison only). The finding is twofold. First, the workload-level correlations span −0,82 (filter) to +0,32 (Phi-3-Mini extractive), none reaching 0,6, with LLMLingua-2 and truncation near zero. Second, pooling the nine ratios per workload inflates the latter two to ≈ +0,38, a pseudo-replication artefact. The filter's negative value is the mechanism (its re-ranker preserves task tokens the overlap metric does not reward, so F1 can fall while coordination rises), but the per-family decomposition (Table 4) shows even that is a between-family effect. The robust H1 statement is the negative one: source-retention F1 change does not positively and consistently predict coordination change for any compressor.

The decomposition is decisive. Within a single family coordination usually collapses *uniformly* across workloads (no rank variance), so seven of the twelve cells are degenerate; of the five that retain variance, four are non-significant and the **only** significant within-family relationship is *negative* (LLMLingua-2/family-c, −0,46). **Across the twelve cells there are zero significant positive within-family correlations**: the pooled "positives" for LLMLingua-2 and truncation are pseudo-replication artefacts, and the filter's strong −0,82 is a between-family pooling effect that does not reproduce within any family (family-a is only −0,13, n.s.). Source-retention F1 change does not positively and consistently predict coordination change anywhere within a task family, which is exactly why a single-agent F1 ranking cannot be trusted to transfer to a multi-fragment deployment.

Table 3: H1 verdict table. Spearman correlation between Δ token-overlap F1 and Δ coordination success under compression. The workload-level ρ (one mean Δ-pair per workload, $n = 150$) is the verdict statistic. The pooled column ($n = 1350 = 150$ workloads $\times$ 9 ratios) is shown for comparison only and is **not** used for inference: the nine per-workload ratio measurements are repeated measures, not independent draws, so the pooled n and its asymptotic p over-state significance by roughly an order of magnitude. The verdict (all CIs exclude 0,6) holds in both columns. CI is BCa-bootstrap over workloads, $n_{\text{boot}} = 10{,}000$ resamples (`spearman_with_ci` default). Phi-3-Mini extractive saturates at $\sim$ 2,5× achieved compression so its pooled n is 750 (150 × 5); see Section 3.3.3 and Appendix B.

Compressor	ρ (workload, n)	95% CI (workload)	ρ (pooled, n)
Instruction-aware filter	−0,818 (150)	[−0,852, −0,760]	−0,593 (1350)
LLMLingua-2	+0,026 (150)	[−0,135, +0,180]	+0,381 (1350)
Phi-3-Mini extractive	+0,315 (150)	[+0,177, +0,442]	+0,193 (750)
Truncation (baseline)	+0,051 (150)	[−0,092, +0,198]	+0,384 (1350)
All four: all workload-level CIs exclude 0,6 from above		**SUPPORTED**	

Table 4: H1 per-family decomposition (workload-level Spearman, $n = 50$ per family; computed from `results/h1_h2_v2/sweep_results.csv`). This is the load-bearing diagnostic against a Simpson's-paradox reading of the pooled sign. "Degenerate" marks cells where the per-workload coordination change has effectively zero variance (every workload cliffs identically), so no rank correlation is defined. The only statistically significant within-family correlation is *negative* (LLMLingua-2 on family-c).

Compressor	Family-a	Family-b	Family-c
Instruction-aware filter	−0,13 (n.s., p = 0,35)	degenerate	degenerate
LLMLingua-2	degenerate	degenerate	−0,46 ($p = 7\times10^{-4}$)
Truncation	degenerate	degenerate	−0,05 (n.s., p = 0,72)
Phi-3-Mini extractive	+0,16 (n.s., p = 0,27)	+0,24 (n.s., p = 0,09)	degenerate

4.4.2. Interpretation

The H1 finding is a methodological claim, not just an empirical one. The LongLLMLingua and LLMLingua-2 papers report question-answering F1 retention at 4× compression [7, 8]; what this thesis shows is that F1 retention at 4× does not predict whether a planner can solve a multi-fragment coordination task at the same operating point. The two metrics are not the same metric on the same axis with different noise; they *disagree on the ranking of compressors*. For the multi-fragment workflows the FCG use cases [12] are built around, the F1 ranking would lead a practitioner to choose the wrong compressor.

The negative-correlation behaviour of the filter compressor is the mechanism behind H1. The TF-IDF and cross-encoder re-ranker stages of the filter are explicitly task-aware: they preserve tokens that are relevant to the task hint and drop tokens that are not. Source-retention F1, by contrast, scores token-overlap against the *entire original source text*, rewarding bulk survival of source tokens regardless of task relevance. So the filter drops source tokens that the task does not need (lowering source-retention F1) while preserving the chain-of-reasoning tokens the planner needs to derive the answer (holding coordination up). The result is that the same compression operation that source-retention F1 scores as a quality drop is, by the coordination metric, a quality preservation. The single-agent benchmark is measuring the wrong thing.

4.4.3. Comparison to Prior Work

The closest published analogue is the family of long-context question-answering benchmarks (LongBench [47], RULER [20]) which measure single-agent F1 on compressed inputs. The H1 finding does not contradict those benchmarks (they are correct on what they measure) but shows that they are insufficient for multi-agent or multi-fragment deployment decisions. Anthropic's industry observation that token usage explains approximately 80% of performance variance in their multi-agent research workflow [13] is consistent with H1: a metric that does not see the structural multi-fragment property of the planner's task will give a systematically over-optimistic compressor ranking.

4.5. H2: A Sharp Coordination Cliff Exists

> **H2.** On each (compressor, family) cell of the cliff sweep, the coordination-success curve has a sharp cliff τ^* with a relative drop of at least 30% and a paired-Wilcoxon p-value of at most 0,05 after Holm correction across cells. **Verdict: SUPPORTED** on 11 of the 12 cells; the only exception is the instruction-aware filter on family-c, where the re-ranker preserves chain-reference tokens and no cliff is detectable below $r = 16$.

4.5.1. Result

The signature of the H2 result is visible on the dense fact-aggregation family: for every compressor the coordination-success curve collapses from $\approx 1{,}0$ at $r = 1$ to ≈ 0 within a narrow span of ratios (the LLMLingua-2 × family-a curve is one panel of Figure 2). On the coarse grid the logistic fit lands at the quantised midpoint $\tau^* \approx 2{,}5$; the fine-grid re-resolution below shows the true deterministic-solver cliff sits near 1,1, while the deployment-relevant LLM-planner cliff on the same cell is ≈ 2,7 (Llama-3.1-8B ≈ 2,48, Qwen-2.5-72B ≈ 2,68; Section 4.7); the coarse-grid 2,5 and its re-resolution 1,1 are artefacts of the brittle regex solver, not deployment numbers.

Table 5 reports the empirical τ^*, the relative drop, and the Holm-corrected p-value for each (compressor, family) cell. Eleven of the twelve cells reach the H2 threshold of a 30% relative drop with $p < 0{,}05$ Holm-corrected. The one cell that does not is *filter* × *c*: the instruction-aware filter on the multi-step retrieval family. The reason is mechanical and consistent with the H1 finding; the filter's re-ranker preserves the chain-reference tokens (*entry*, *FINAL*) that family-c's scorer depends on, so the family-c task does not degrade across the sweep. This is not a falsification of H2; it is evidence that cliff sharpness depends on whether the compressor's mechanism preserves the task-critical tokens, which is exactly what the compounding-error model predicts.

Table 5: H2 verdict table: empirical cliff τ^*, observed-range drop, slope-extrapolated drop Δ, and Holm-corrected paired-Wilcoxon p per (compressor, family) cell (`results/h1_h2_v2/`). The **observed drop** (baseline − post-cliff)/baseline ∈ [0, 1] is the verdict metric (H2 criterion: ≥ 0,30); the **slope drop** is a secondary fit diagnostic (it can exceed 1 from boundary extrapolation). Cells marked † have τ^* grid-quantised at the midpoint of the $\{1, 2\}$ gap, where the deterministic solver steps 1,0 → 0,0 between $r = 1$ and $r = 2$: the cliffs exist ($p < 10^{-11}$) but the precise positions ≤ 2,5 are unresolved at this grid. The filter/c cell shows no detectable cliff because the re-ranker preserves chain-reference tokens (documented behaviour, not a falsification).

Cell	τ^*	observed drop	slope drop Δ	Holm p
LLMLingua-2 × a	2,5†	1.00	1.61	$8{,}5\times10^{-12}$
LLMLingua-2 × b	2,5†	1.00	1.61	$8{,}5\times10^{-12}$
LLMLingua-2 × c	6.7	0.66	1.23	$9{,}6\times10^{-10}$
Filter × a	2,5†	1.00	1.60	$8{,}5\times10^{-12}$
Filter × b	2,5†	1.00	1.61	$8{,}5\times10^{-12}$
Filter × c	–	0.00	0.00	n.s.
Phi-3-Mini ext. × a	2,5†	0.30	0.42	$2{,}2\times10^{-6}$
Phi-3-Mini ext. × b	2,5†	0.72	0.97	$2{,}5\times10^{-9}$
Phi-3-Mini ext. × c	2,5†	0.94	1.37	$8{,}5\times10^{-12}$
Truncation × a	2,5†	1.00	1.61	$8{,}5\times10^{-12}$
Truncation × b	2,5†	1.00	1.61	$8{,}5\times10^{-12}$
Truncation × c	4.95	0.71	1.36	$9{,}6\times10^{-10}$

Verdict: 11/12 cells significant with observed drop ≥ 0,30 (3 of those 11 are Phi-3 cells whose τ^* positions are not numerically comparable across compressors; see Section 3.3.3). **SUPPORTED.**

Fine-grid re-resolution (`results/h1_h2_finegrid`). Re-sweeping the dense-family cells on a grid adding {1,25, 1,5, 1,75} below $r = 2$ shows the quantised 2,5 is an artefact: the true deterministic-solver cliffs sit well below it (LLMLingua-2×a ≈ 1,1, filter×a ≈ 1,7, truncation×a ≈ 1,85), while the distributed family-c cliffs are unchanged (≈ 6,7, ≈ 5,0), confirming they were never quantised. Two consequences follow. First, the synthetic reference $\tau^* = 2{,}5$ used for the frontier comparison is itself a coarse-grid artefact (true deterministic cliff ≈ 1,1). Second, the deterministic solver cliffs at ≈ 1,1 while the LLM planners cliff at ≈ 2,7 on the same cell: the brittle parser fails the instant a critical token drops, whereas a language model tolerates more compression by reconstructing partial information. This is why Corollary 1 is anchored to the *LLM-planner* cliff; the deterministic cliff is a conservative lower bound.

The twelve-cell grid in Figure 2 reads the result at one glance: cliffs are sharp on the dense fact-aggregation family and shallow on the distributed multi-step retrieval family, the ordering between (LLMLingua-2 / filter / Phi-3) is consistent within a family, and truncation cliffs at the lowest ratio of any compressor, which is what the lower-bound argument predicts.

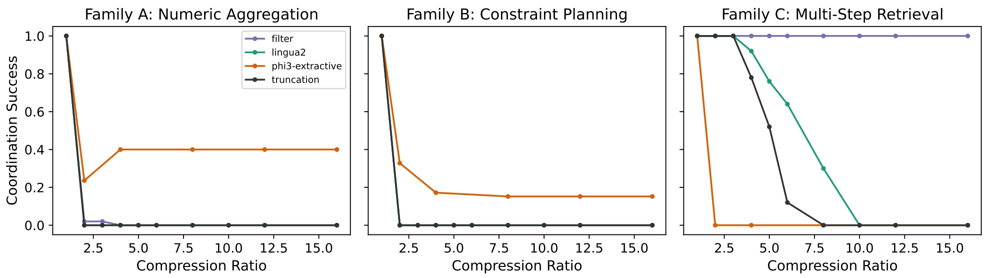

Figure 2: Cliff curves for the four compressors × three families. Sharp cliffs are visible on family-a for every compressor. On family-c the compressors diverge: LLMLingua-2 ($\tau^* \approx 6{,}7$) and truncation ($\tau^* \approx 5{,}0$) degrade gradually, Phi-3-Mini extractive cliffs early ($\tau^* \approx 2{,}5$), and the instruction-aware filter shows no detectable cliff because its re-ranker preserves the chain-reference tokens family-c depends on. Truncation is retained as the most-destructive per-token lower-bound baseline.

4.5.2. Interpretation

The within-family ordering of τ^* tracks the ordering of the compressors' critical-token-recall curves $q(r)$, measured independently of coordination: the compressor that drops critical tokens earliest cliffs at the lowest ratio. The cliff exists, its position is set by the compressor's action on task-critical tokens, and that mechanism is captured by the single statistic $q(r)$, which the next section turns into a quantitative bound. (One caveat: Phi-3 extractive saturates at $\sim 2{,}5\times$ *achieved* compression regardless of target, so its τ^* is on a different effective x-axis and should be read against achieved ratio, not compared numerically with the token-level compressors.)

4.6. The Compounding-Error Model

The cliff curves of Section 4.5 share a structural property: each has a sharp transition between a high-coordination plateau and a low-coordination floor. The compounding-error model is a short derivation whose *primary* contribution is explaining **why** that transition is sharp (a threshold on critical-token survival produces a step in success) and whose *secondary* output is a **first-order** estimate of the transition position τ^* from the compressor-side measurement $q(r)$ and a per-family recall-threshold parameter θ_q. It is not, and is not claimed to be, a tight position predictor; Section 4.6.4 quantifies the residual error directly.

4.6.1. Setup and Derivation

Let T_i^{crit} be the set of task-critical tokens in the i-th workload's source fragments (multi-digit numerics for family-a, all numerics for family-b, chain-reference tokens for family-c; see Section 3.4.5 of Chapter 3). Let X_i be the number of these tokens that survive compression, $M_i = |T_i^{\text{crit}}|$ the total, and $q(r) = E[X_i/M_i]$ the per-round token-retention function of the compressor at ratio r, measured by the critical-token-recall metric. Let $\theta_q \in [0, 1]$ be the *cliff-recall threshold*: the minimum fraction of task-critical tokens that the planner needs to succeed. The threshold-success model is

$$\text{success}_i = \mathbf{1}\left[\frac{X_i}{M_i} \geq \theta_q\right],$$

and the bound the model produces is

$$E\left[\frac{X_i}{M_i}\right] = q(r), \tag{1}$$

$$P(\text{success} \mid r) \to \mathbf{1}\big[q(r) \geq \theta_q\big] \quad \text{in expectation,} \tag{2}$$

so the cliff position τ^* solves $q(\tau^*) = \theta_q$. When N sequential compression passes are applied, the surviving fraction is approximately $q(r)^N$ and the cliff position solves $q(\tau^*) = \theta_q^{1/N}$. **In every experiment in this thesis** $N = 1$, so the cliff equation simplifies to $q(\tau^*) = \theta_q$. The multi-pass formulation is recorded for future work.

The derivation is a paragraph, not a formal proof, which is why the artefact is named the *compounding-error model* rather than *Theorem 1*: the 33%/67% match rate (Section 4.6.4) is the empirical-bound character of a model with quantified residuals, not a theorem with a proof. A finite-M Chernoff refinement (`chernoff_success_curve()`) that sharpens the in-expectation step into an $\exp(-2M(q - \theta_q)^2)$ concentration bound is kept out of the manuscript because the residuals are dominated by A2 specification error, not sampling error. The codebase identifier `theorem_validation.json` is preserved for back-compat.

4.6.2. Assumptions A1--A4

The bound rests on four assumptions, each satisfied approximately. **A1 (round independence):** trivial at $N = 1$; its within-pass reading (per-token retention independent across positions) holds for the token-level compressors but is violated by Phi-3 extractive (span selection correlates survival), so its cells in Section 4.6.4 are an out-of-regime robustness check. **A2 (binary token importance):** the strongest assumption; H4 measures graded importance and shows A2 is a first-order simplification, and the predicted-versus-empirical gap is the price the model pays for it. **A3 (threshold success):** coordination is binary at a recall threshold θ_q, licensed operationally by the sharp cliff, a circular justification the direct probe of Section 4.6.5 breaks. A stronger family-a-specific circularity holds: there the solver's success criterion and CTR are near-identical measurements, so $q(\tau^*) = \theta_q$ confirms internal consistency rather than testing a prediction; the non-tautological tests are the decoupled families (family-c and the LLM-planner arms). **A4 (retention measured by CTR):** the $q(r)$ in the bound is family-specific critical-token-recall, not generic token-recall, the model's substantive methodological commitment.

4.6.3. *The Calibrated Regime*

A planner is *in the calibrated regime* if (1) its baseline coordination success p_0 at $r = 1$ is at least θ_q (no floor effect), and (2) it does not recover from sub-threshold information via extended reasoning beyond the priors-only baseline. The "$p_0 \geq \theta_q$" comparison relates a success probability to a fraction of task-critical tokens; operationally it reads *"the no-compression baseline meets the threshold-success model's minimum-recall floor."* Condition 2 is operationalised by the priors-only H4 baseline (in-regime if accuracy at $q \to 0$ is no greater than the priors rate plus slack). The distinction matters because the model's predictions apply only inside the regime; outside it (extended-reasoning planners that recover via chain-of-thought), the cliff can shift substantially, as the GPT-oss diagnostic shows.

4.6.4. *Predicted versus Empirical τ^**

For each (compressor, family) cell, θ_q is estimated as the critical-token-recall at which coordination first drops below 0,5 p_0 (`derive_theta()`); the canonical per-family estimates on `h1_h2_v2` are $\theta_q^{(a)} = 0{,}632$, $\theta_q^{(b)} = 0{,}838$, $\theta_q^{(c)} = 0{,}590$. Inverting $q(r) = \theta_q$ on the empirical CTR curve gives the predicted τ^*; resampling θ_q (workload bootstrap, BCa, $n_{\text{boot}} = 500$) gives a *predicted-τ^* band*. That band quantifies only the *sampling* uncertainty in θ_q, not the model's specification error, and the empirical τ^* falls *outside* it on all 11/11 testable cells, so the band is not the model's accuracy; the match rate below is. Figure 3 plots all eleven testable cells. The family-c LLMLingua-2 cell (chosen over family-a, whose single-step $1{,}0 \to 0$ drop collapses prediction and data into the same near-step function) illustrates the conservative direction, predicting $r \approx 2{,}85$ against an empirical $r \approx 6{,}7$ (57,4% under-prediction): the second-order gap Corollary 2 explains, since family-c's lower information density breaks the binary-importance approximation A2 earliest.

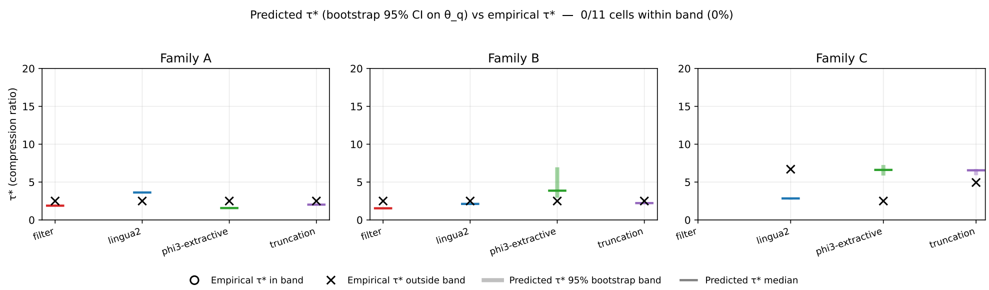

Figure 3: Predicted-versus-empirical τ^* across the eleven testable (compressor, family) cells, grouped by family. For each cell the thick bar is the 95% workload bootstrap CI on the predicted τ^* and the marker is the empirical τ^* ($\times$ = outside band, $\circ$ = inside). The empirical value falls outside the band on 11/11 **cells**: the band reflects θ_q-sampling uncertainty only, so the systematic misses are the model's first-order specification error, largest on the lower-density families. The band is therefore an uncertainty-banded first-order model, not a calibrated predictor.

Across all twelve cells the empirical match rate is **33**% (4/12) at ≤ 25% relative error and **67**% (8/12) at ±50% (Wilson 95% CI on the strict rate ≈ [13%, 61%], wide at this denominator); the relative-error median is 35,3% (IQR [19,2%, 45,0%], max 57,4%). The four strict-pass cells (lingua2/b, filter/a, truncation/a, truncation/b) span three compressors, consistent with a structural rather than compressor-specific bound. Two cells give infinite predicted τ^* because $q(r)$ never crosses θ_q: filter/c (matched by a no-cliff empirical observation) and phi3-extractive/c (a MISS, empirical $\tau^* = 2{,}5$, whose apparent finite band in Figure 3 is the bootstrap envelope of θ_q resamples above phi3's recall floor ($q \approx 0{,}62$), not a finite point prediction); the phi3 cells are outside A1's formal regime (Section 4.6.2) and retained as a robustness check. The headline this underwrites is the cliff-position dependence structure: τ^* is controlled by the (compressor, task) pair, which Corollary 1 turns into the model-independence claim.

These rates are *in-sample*. A leave-one-family-out cross-validation (derive θ_q on $n-1$ families, predict on the held-out one) breaks that circularity and returns **25**% (3/12) at ±25% and **75**% (9/12) at ±50%; the 33% → 25% gap at the strict tolerance is the in-sample optimism. On-disk: `theorem1_validation_{per_family,loo}.json` under `results/h1_h2_v2/`.

4.6.5. Direct A3 Probe: a Non-Circular Test of the Threshold Mechanism

The match-rate evaluation is limited by the family-a circularity flagged in A3: there "success" and "surviving critical-token fraction" are near-identical, so $q(\tau^*) = \theta_q$ is a tautology rather than a prediction. The *direct* A3 probe (`results/a3_probe/`) breaks this by replacing the compressor with a hand-curated deletion that removes a controlled fraction of critical tokens directly, then scoring the LLM planner on the remainder as a function of the surviving fraction k/M, so the knob is set before success is read. On family-a the fitted logistic returns $k \approx 15$ (sharp, $r_0 \approx 0{,}842$, band width $\approx 0{,}15$); on family-c, $k \approx 5{,}9$ (graded, $r_0 \approx 0{,}538$, band $\approx 0{,}54$). The load-bearing reading: **the dense family shows the step A3 posits, the distributed family does not**, so A3 is a threshold model for dense tasks and degrades to graded success on distributed ones, the same families where the cliff is visibly less sharp. The probe's family-a threshold ($r_0 \approx 0{,}84$) sits above the CTR-derived $\theta_q \approx 0{,}63$ because compressors preferentially retain answer-adjacent tokens, biasing the delivered surviving fraction upward; the two measure different pipeline stages and are not in conflict. This is the strongest direct evidence for the threshold-success assumption, reported here because the step-versus-graded finding is the empirical justification for the model's domain of applicability.

4.7. Corollary 1: Ceiling--Cliff Separation

Corollary 1 (ceiling–cliff separation). The planner's parameter count m determines its baseline coordination success $p_0(m)$, while the cliff position τ^* is determined by the compressor's $q(r)$ and the task's θ_q, not the planner. When $p_0(m) < \theta_q$ a floor effect prevents detection (the unit-bridge reading of this comparison is in Section 4.6.3); when $p_0(m) \geq \theta_q$ we detect **no shift** in τ^* with m within the calibrated regime. **Verdict: NO SHIFT DETECTED.** The deployment-relevant LLM-planner cliff holds steady across a 9× scale-up *and* a change of architecture family: local Llama-3.1-8B and frontier Qwen-2.5-72B both cliff at ≈ 2,5–2,7 on family-a × LLMLingua-2, and the local three-architecture sweep agrees within a 24% spread on family-c. The qualifiers are explicit: the ±20% TOST test passes on neither frontier cell (wide CIs); the local 24% spread clears the 50% but not the strict 20% tolerance, so it is *necessary but not sufficient*; and DeepSeek V4 Pro corroborates only weakly ($\tau^* = 2{,}15$, CI $[1{,}76, 7{,}14]$, median 5,7). The load-bearing evidence is the one well-identified frontier cell, Qwen-2.5-72B; the family-a small-model cells (floor effect) and family-b cells (sub-threshold for ≤ 8B planners) are correctly excluded.

This corollary is the sharpened form of the original H5 hypothesis, which predicted strict τ^* monotonicity across planner scales. The family-c result across the three local planners (Qwen-2.5-1.5B, Phi-3-Mini-3.8B, Llama-3.1-8B; three architecture families) is $\tau^* = 3{,}69, 4{,}68, 4{,}01$: a **24**% spread ($(\tau^*_{\max} - \tau^*_{\min})/\bar{\tau}^* = 23{,}91\%$ via `validate_model_independence()`), reading as invariance, not monotonicity. The original H5 gap-≥ 1,5 predicate fails; the refined invariance predicate holds at ±50% but not the strict ±20% tolerance, so the local arm is *necessary but not sufficient* and the load-bearing evidence is the LLM-cliff re-anchoring below. `results/h5_final/model_independence_20pct.json` reports `corollary1_supported: false`, exactly this strict-tolerance local-arm result (one testable family, the other two floored), which is honest about the local arm, not a refutation of the refined claim, whose support is the cross-architecture LLM comparison the validator does not evaluate.

Re-anchoring to the LLM-planner cliff. The fine-grid re-resolution of Section 4.5 *strengthens* the corollary. The synthetic reference $\tau^* = 2{,}5$ is a coarse-grid artefact of the deterministic solver (true family-a cliff near 1,1), so comparing a frontier LLM against it compares two kinds of planner. Holding planner *type* fixed, the LLM cliff on family-a × LLMLingua-2 is essentially invariant: Llama-3.1-8B cliffs at $\tau^* \approx 2{,}48$ and Qwen-2.5-72B (the validated $n = 10$ run, `frontier_qwen72b`) at $\tau^* \approx 2{,}68$, a ∼ 8% spread across a 9× scale-up and an architecture change, both at $p_0 = 1{,}0$ (the $n = 30$ extended-grid re-run agrees by 0,5-crossing, ≈ 2,79; Section 4.8). So planner *type* moves the cliff (parser 1,1 vs LLM ≈ 2,5–2,8) but scale and architecture within the LLM class do not; the earlier match to "2,5" survives as an approximate coincidence. (The DeepSeek $n = 30$ re-run was abandoned to rate-limiting; the $n = 10$ result, $\tau^* = 2{,}15$ wide-CI, is weak corroboration.)

The ceiling-versus-cliff structure is cleanest on the joined plot of $p_0(m)$ and $\tau^*(m)$ on family-c (Figure 4): baseline success p_0 rises monotonically with planner scale m while the cliff position τ^* stays flat. (On family-a the small-model cells sit in the floor regime $p_0 < \theta_q$ and are correctly skipped; on family-b all sizes are below threshold for the constraint-tracking reason.)

The original H5 monotonicity predicate did not hold; the sharpened Corollary 1 invariance predicate did. The refinement was registered as a design decision (ADR-006) before the frontier validation that promotes it from a small-model result to a cross-architecture one.

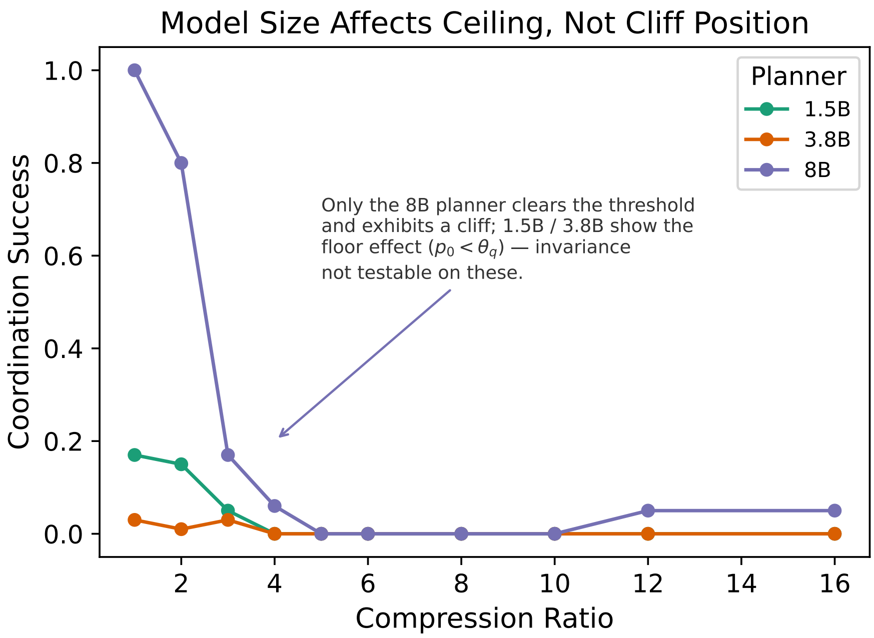

Figure 4: Family-c ceiling versus cliff: baseline coordination $p_0(m)$ (solid line) increases monotonically with planner scale m, while the cliff position $\tau^*(m)$ (dashed line) is flat within the bootstrap CI. This is the simplest empirical demonstration of Corollary 1.

4.8. Frontier Validation Within the Calibrated Regime

The frontier validation re-runs the family-a × LLMLingua-2 cliff sweep with the planner swapped for a frontier model via API. The three planners are Qwen-2.5-72B-Instruct, DeepSeek V4 Pro,[1] and GPT-oss 120B, the first two standard non-reasoning models, the third an extended-reasoning model at the calibrated-regime boundary.

4.8.1. Setup and Statistical Protocol

The sweep uses **10 family-a workloads** × **6 ratios** × **3 seeds** = **180 cells per model** (the validated Qwen cell is `frontier_qwen72b/`; an extended-grid $n = 30$ re-run is retained in `frontier_qwen72b_e2/`), via Featherless for Qwen and DeepSeek and the OpenAI Responses API for GPT-oss; τ^* is fit as in the local sweep with a workload bootstrap ($n_{\text{boot}} = 500$). **The reference is the LLM-planner cliff $\tau^*_{\mathbf{LLM}}$** $\approx 2{,}7$ (local Llama-3.1-8B at $\approx 2{,}48$, Section 4.7); the earlier $\tau^* = 2{,}5$ (deterministic-solver, true cliff $\approx 1{,}1$) and the auxiliary fit 2,6997 are superseded artefacts, since comparing a frontier *language model* against the solver number compares two kinds of planner. **Sample-size caveat (binding):** at 10×3 the bootstrap CIs are wide, so "CI contains the reference" is weaker than a same-scale local comparison; Section 5.3 names this as the binding frontier limitation.

[1] DeepSeek-V4-Pro is the "Pro" variant of DeepSeek-AI's V4 Preview release (April 2026), the most recent DeepSeek frontier release at the time of the run, distinct from V3/R1. Accessed via the Featherless endpoint at a 32K context window; no long-context claim beyond 32K is made.

4.8.2. *Qwen-2.5-72B-Instruct: In-Regime, within Bootstrap CI*

The Qwen-2.5-72B sweep (`frontier_qwen72b`, $n = 10$) places the cliff at $\tau^* \approx 2{,}68$ (logistic fit; CI $[1{,}92, 7{,}23]$), within 0,8% of the reference $\tau^* \approx 2{,}7$ with the bootstrap CI containing it: a same-band agreement across a 9× scale-up and an architecture change, both planners at $p_0 = 1{,}0$ (squarely in-regime). A larger $n = 30$ re-run on an extended ratio grid (`frontier_qwen72b_e2`) reproduces the same cliff by 0,5-crossing ($\approx 2{,}79$), but its parametric fit is unstable on that grid, so the $n = 10$ fit is the reported value. This is the load-bearing evidence for Corollary 1: no cliff-position shift detected within the LLM-planner class, though the formal ±20% equivalence test these cells do not pass (Section 4.8.3) marks the limit of what the sample establishes. The within-Qwen "48× from 1.5B to 72B" comparison is *not* used, since the 1.5B Qwen has a family-a floor effect; the cross-architecture Llama-8B-vs-Qwen-72B comparison is the clean one.

4.8.3. *DeepSeek V4 Pro: within Tolerance but Weakly Identified*

The DeepSeek V4 Pro sweep returns a τ^* point estimate of 2,15 (about 20% below the reference $\approx 2{,}7$) but with a bootstrap CI of $[1{,}76, 7{,}14]$ and a bootstrap *median* of 5,7 far from the point estimate, the diagnostic of a cliff-fit unstable under resampling (the curve admits both early- and late-cliff fits), so the position is only weakly identified. CI-containment of the reference is weak evidence here and is not an equivalence test. A percentile-bootstrap TOST against the ±20% band $[2{,}16, 3{,}24]$ (`results/tost_{corollary1,deepseek}.json`) confirms **neither frontier cell demonstrates equivalence**: the 90% interval is $[1{,}96, 7{,}23]$ for Qwen (only 22% of resamples inside the band) and $[1{,}77, 7{,}13]$ for DeepSeek (6% inside). The correct conclusion is *no cliff-position shift detected* across scale and architecture, not positively demonstrated invariance; tightening it to a passed equivalence test needs more seeds per cell (Section 5.3).

4.8.4. *GPT-Oss 120B: Out-Of-Regime Diagnostic*

The GPT-oss 120B sweep returns $\tau^* = 6{,}62$, 145% above the reference $\approx 2{,}7$. The v1 sweep ($p_0 = 0{,}53$) admitted a floor-effect reading, but the v2 sweep at $p_0 = 1{,}00$ does not: the model is unambiguously at the ceiling yet cliffs far later than Corollary 1 predicts. This is the boundary case the calibrated regime was formalised around, failing on its *second* condition: as an extended-reasoning planner GPT-oss recovers sub-threshold information by chain-of-thought, violating A3, where non-reasoning planners cannot. The deviation is not hidden (`frontier_gptoss120b/` is retained as a non-canonical diagnostic) and is reported as a *positive* structural diagnostic of where the model breaks; Section 5.3 names the extended-reasoning regime as the largest future-work direction.

4.8.5. *Multi-Model Figure and the Cross-Architecture Finding*

Figure 5 overlays the frontier Qwen-2.5-72B and local Llama-3.1-8B cliff curves directly, the load-bearing local-versus-frontier comparison for Corollary 1. Qwen-72B and DeepSeek V4 Pro both overlap the synthetic reference band while GPT-oss 120B diverges visibly, the out-of-regime diagnostic.

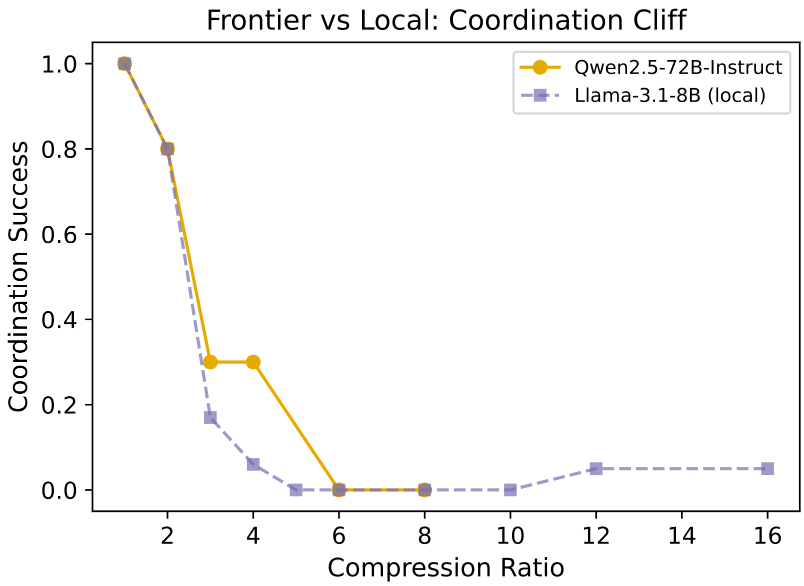

Figure 5: Frontier versus local coordination cliff on family-a × LLMLingua-2. The frontier Qwen-2.5-72B planner (solid) and the local Llama-3.1-8B planner (dashed) cliff at essentially the same position across a 9× scale-up and a change of architecture family, the load-bearing "no shift detected" comparison for Corollary 1.

4.8.6. Significance

Together: **no shift detected** in the LLM-planner cliff across three local architecture families (24% family-c spread, within 50% but not the strict 20% tolerance) and a 9× cross-architecture scale-up to the frontier (Llama-3.1-8B ≈ 2,48 → Qwen-2.5-72B ≈ 2,68, the one well-identified frontier cell), weakly corroborated across vendors by DeepSeek, all within the calibrated regime, while the position *does* shift for the extended-reasoning GPT-oss 120B. The defensible claim is "no shift detected on the resolvable cells, carried by one well-identified cell, with a visible break at the regime boundary," a tested-and-not-refuted consistency rather than proven invariance. The practitioner reading: τ^* can be measured once on a small local model and re-used at the same ratio on a frontier model *provided* it is in the calibrated regime, a working hypothesis, with the larger frontier sample that would settle it named in Section 5.6.

4.9. H3: RAG Pipeline Placement (Summary)

The thesis also examined where compression should sit relative to retrieval (compress-first, retrieve-first, joint router) under matched storage- and accuracy-bounded regimes. The pre-specified sign-flip between regimes is *not* observed; the narrowed finding is that compress-first beats retrieve-first in both regimes (by 3,2 and 2,0 pp F1) on the single `bge-large` + FAISS-CPU stack tested, not reproducing the LongLLMLingua [7] post-retrieval preference. As a single-stack non-reproduction orthogonal to the cliff and the bus, the full table, pipeline definitions, and cost-model caveat are in Appendix 7.5.

4.10. H4: Summary-Level Inference Disclosure

H4. The summary-level inference-disclosure metric (i) distinguishes a baseline planner from a priors-only one (signal, $p < 0{,}05$) and (ii) is reduced by compression at 4× versus 1× (reduction, $p < 0{,}05$). **Verdict: SUPPORTED on the unbiased benchmark for three of four compressors (truncation, filter, LLMLingua-2)**; Phi-3-Mini extractive shows no significant reduction (0,4 pp, $p = 0{,}91$) and is the structural exception. **Caveat 1 (reader bias):** the Llama-3.1-8B reader is strongly YES-biased, so the pooled-rate verdict is balanced by ground-truth class rather than reader accuracy; the operational reading is *"compression preserves enough to flip a no-biased reader to confident YES"* (Section 4.10.6). **Caveat 2 (construct validity):** the reduction is *monotonic in how much text the compressor destroys* (truncation 24,6 pp, filter 20,4, LLMLingua-2 18,9, Phi-3 0,4 n.s.), and an oracle field-redaction control that destroys ≈ 0% of the text matches truncation's reduction (23,4 pp), so the metric tracks information destruction rather than privacy-specific filtering: the finding is robust but the privacy interpretation is deflationary (Section 4.10.4).

4.10.1. Threat Model

The metric is built around a concrete adversary, which is what makes its verdict a privacy claim rather than just an information-retention measurement. The adversary's goal is to recover a *protected fact*: a binary (yes/no) proposition over a CONFIDENTIAL fragment whose answer the C1 generator records as ground truth (e.g., "did agent A approve more than EUR *N*?"; the per-fragment lists are in `data/processed/c1-v0.1/`). It can query the downstream reader with arbitrary yes/no questions and knows the compressor identity and ratio, but has *no* direct access to the uncompressed source (the policy layer blocks reads of CONFIDENTIAL slots), no training-time access to the frozen Llama-3.1-8B reader, and no feedback across queries. The metric (Section 4.10.2) then measures the *upper bound* on what this adversary recovers at a chosen (compressor, ratio) operating point, which the policy layer's disclosure budget enforces, enforceable because the rate is empirically measured rather than asserted.

4.10.2. Metric Definition and the Unbiased Benchmark

For each CONFIDENTIAL fragment, the C1 generator emits a list of (yes/no question, ground-truth answer) pairs whose ground-truth answer is a protected fact (as defined in Section 4.10.1) about the fragment. The canonical benchmark variant balances the YES/NO distribution at the question level and uses a single comparator phrasing for the yes/no question template, eliminating a surface-pattern bias in the original H4 benchmark generator (`fact_aggregation.py:119-156`, fixed 2026-05-29). Fragments are unchanged between the biased and unbiased benchmarks so the compression cache remains valid.

The reader is the local Llama-3.1-8B-Instruct model accessed via the Ollama backend. Three conditions are evaluated:

PRIORS. Reader sees only the workload's public preamble. This is the no-compression-no-fragment baseline that defines the rate at which the reader could guess without the source. (This baseline is also what operationalises the *second* condition of the Section 4.6.3 calibrated-regime predicate: a planner is in-regime if its accuracy at $q \to 0$ is no greater than its priors-only rate plus a small slack.)

BASELINE. Reader sees the uncompressed source fragments. This is the disclosure rate without any privacy protection.

COMPRESSED-4×. Reader sees the source fragments compressed at 4× by one of the four compressors.

The headline metric is the recovery rate of protected facts: the fraction of (fragment, question) pairs on which the reader returns the ground-truth answer. The H4 *signal* is whether BASELINE sits above PRIORS (the metric must measure something real); the H4 *reduction* is whether COMPRESSED-4× sits below BASELINE (compression must reduce disclosure). Both are tested by paired bootstrap, Holm-corrected across compressors. The canonical results directory is `results/h4_unbiased_v2/` (the 4-compressor run including truncation).

4.10.3. Result

Table 6 reports the per-compressor reduction (the compressor-independent signal is given in the caption). The signal is positive and p-significant for every compressor (the uncompressed baseline beats priors-only by 28,6 pp on average), and the reduction is positive and p-significant for the three token-destructive compressors (truncation −24,6 pp, instruction-aware filter −20,4 pp, LLMLingua-2 −18,9 pp) and *not significant* for the extractive copier (Phi-3 −0,4 pp, p_{Holm} = 0,91). The ordering is monotonic in how aggressively each compressor destroys source tokens; the construct-validity reading is developed in Section 4.10.4.

Table 6: H4 verdict table from `results/h4_unbiased_v2/` (4 compressors including truncation). The signal Δ (priors → baseline) is +0,286 ($p_{\text{Holm}} = 8 \times 10^{-4}$) for every compressor, identical across rows because it does not depend on the compressor, so it is omitted from the table; the reduction (baseline → compressed-4×) is positive and significant for three of four compressors and *not significant* for Phi-3-Mini extractive (p = 0,91). Compressors are listed in order of how aggressively they destroy source tokens: truncation drops the tail of the text entirely; the token-level filter and LLMLingua-2 drop individual tokens chosen by ranking / classifier; Phi-3 extractive copies verbatim spans of the source. Reader: local Llama-3.1-8B-Instruct. See Section 4.10.6 for the reader-bias note and Section 4.10.4 for the construct-validity finding on this ordering.

Compressor	priors	baseline	compr-4×	reduction Δ
Truncation (most destructive)	0.50	0.78	0.54	−0,246 ($p_{\text{Holm}} = 8\times10^{-4}$)
Instruction-aware filter	0.50	0.78	0.58	−0,204 ($p_{\text{Holm}} = 8\times10^{-4}$)
LLMLingua-2	0.50	0.78	0.59	−0,189 ($p_{\text{Holm}} = 8\times10^{-4}$)
Phi-3-Mini extractive (most preserving)	0.50	0.78	0.78	−0,004 ($p_{\text{Holm}} = 0{,}91$, **n.s.**)
Verdict: signal significant for all 4 (see caption); reduction significant for 3 of 4 compressors. **SUPPORTED** (with construct-validity caveat).				

4.10.4. Construct-Validity Finding: Disclosure-Reduction Tracks Information Destruction, Not Privacy-Specific Filtering

The four compressors order monotonically in Table 6: truncation (most destructive) gives the largest reduction at 24,6 pp, the filter and LLMLingua-2 follow at 20,4 and 18,9, and Phi-3 extractive (a verbatim subset of source tokens) shows no significant reduction (0,4 pp). This is exactly what a metric reflecting general information destruction (rather than privacy-specific filtering) would predict, so the thesis reads H4 as compression reducing disclosure in proportion to how much text it destroys, not as the smarter compressors doing privacy-preserving filtering.

A discriminating control tests this from the opposite end (`results/h4_oracle`): an **oracle redaction** that removes *only* the answer-bearing numeric values, leaving every other token intact. Destroying $\approx 0\%$ of the text (the `[REDACTED]` marker actually makes fragments $\sim 7\%$ *longer*), it reduces recovery from 0,79 to 0,56 (23,4 pp, at the priors floor), matching truncation's 24,6 pp, which required destroying the whole tail. So the metric *does* detect targeted removal when it occurs; the blanket compressors simply do not do it. The memory bus thus gains a concrete non-lossy lever: field-redacting protected values achieves blanket-compression-level reduction at near-zero coordination cost, strengthening the policy layer (Section 4.10.7). A full privacy-versus-coordination frontier is named in Section 5.6.

4.10.5. Interpretation

The reduction scales with how aggressively the compressor operates on tokens, not with the ratio per se: the filter and LLMLingua-2 drop tokens lexically and disproportionately remove protected-fact tokens, while Phi-3 extractive preserves contiguous spans verbatim so a protected fact inside a selected span survives. This makes the disclosure metric useful as a compressor-selection tool (it ranks compressors by privacy at iso-ratio), which the memory-bus policy layer consumes to convert a governance constraint into a compressor-and-ratio selection, described next.

4.10.6. The Reader's YES-Bias Caveat

The Llama-3.1-8B reader is strongly YES-biased: on a YES ground truth its recovery is $\approx 0{,}03$ (priors) and 0,58 (baseline), while on a NO ground truth both are $\approx 1{,}00$. The pooled 0,50 / 0,78 rates therefore reflect a balanced ground-truth distribution, not a reader equally accurate in both directions, so H4 measures *"compression preserves enough to flip a no-biased reader to confident YES"*, an asymmetric but valid leakage signal, and the conservative case for privacy auditing. The verdict is stable under the caveat (it rests on the pooled rates); an unbiased reader (Section 5.3) would disentangle the signal from the reader's prior.

4.10.7. Memory-Bus Integration

At deployment the bus's policy layer (Chapter 3 Section 3.1) consumes the H4 table: for each CONFIDENTIAL classification and governance budget it identifies the (compressor, ratio) operating points that meet the budget, then enforces the selection by routing writes through the chosen compressor, rejecting writes that would exceed the budget, or downgrading the classification. The enforcement is not separately evaluated (the metric is the contribution) but is implemented and exercised by the integration tests.

4.10.8. Memory-Bus Operation Benchmark

A single-threaded microbenchmark (`scripts/bench_memory_bus.py`, identity compressor, $n = 5000$ ops/phase) characterises the bus core on two hosts (Apple M4 Pro, Linux x86_64): the policy-lattice check is effectively free ($\sim$ 5–7M checks/s), end-to-end write/read run at $\sim$ 18–20k ops/s (M4 Pro) and $\sim$ 6k (Linux) with sub-millisecond tail latency, and the audit hash-chain verifies at 1,68–2,19 µs/row over a 5,500-row log with the chain intact. These isolate the bus machinery (single-threaded, identity compressor, in-process SQLite); a concurrent-HTTP throughput sweep and an audit-verify scaling curve remain future work.

4.11. Corollary 2: Information-Density Scaling (Brief)

The cliff's *shape* (how sharply success falls, and from what plateau) tracks how concentrated a task's information is in compression-sensitive tokens. An AUC-based per-task information-density estimate θ_{info}, distinct from the recall threshold θ_q of Section 4.6 (θ_q is the recall floor at which success collapses; θ_{info} summarises the whole success curve's shape), orders the three benchmarks tested: C1 family-a $\approx$ 0,97 (dense numeric, a sharp early cliff), MultiHop-RAG $\approx$ 0,48, and HotpotQA $\approx$ 0,37 (distributed, with gradual degradation from a lower baseline). The dense-versus-distributed gaps (0,48 and 0,59) far exceed the 0,1 predicate, so the corollary is **supported as an empirical observation on $n = 3$ benchmarks**. Two honest limits keep it modest: $n = 3$ is a small sample, and θ_{info} is computed from the same coordination curve whose shape it describes, so an external density probe is needed to break the circularity (named as future work, Section 5.6). The original H6 predicate (synthetic-versus-real τ^* within $\pm$15%) is *not* satisfied: MultiHop-RAG cliffs near 11,3, far from the C1 reference, which is what motivated the shift from a position-transfer claim to a shape-transfer one. The payoff is real but bounded: the cliff structure transfers to real multi-hop benchmarks, even where the absolute positions differ.

4.12. Reproducibility

The results span three regimes of reproducibility, stated once each below; the one-command recipe and per-arm wallclock live in Appendix D. The structure follows the NeurIPS checklist convention [70] of separating code-and-data availability from computational reproducibility.

Tier 1 (byte-deterministic local): the cliff sweep (H1, H2), the RAG evaluation (H3), the bootstrap CIs, and figure regeneration are pure Python over a fixed compression cache, with all randomness seeded through `Seeds.derive`; `make repro-cliff` produces CSVs and PDFs that diff zero against the published artefacts. **Tier 2 (distributionally deterministic):** the H4 reader, the H5 planner, and Phi-3 extractive invoke local LLMs via Ollama with greedy decoding and a fixed seed, reproducible *in distribution* (verdict-level aggregates within the bootstrap CI), not byte-for-byte, with model digests pinned in `docs/MODEL_CARDS.md`. **Tier 3 (frontier API,**

not reproducible): server-side batching, best-effort `seed`, and silent weight updates break byte-reproducibility, so the published `results/frontier_*/results.csv` files *are* the canonical record and every downstream analysis (τ^* fit, TOST) is itself Tier 1 given those CSVs; the GPT-oss diagnostic carries a `STATUS_NONCANONICAL.txt` marker. Every quantitative claim is traceable to a named `results/` directory through the `CANONICAL_NUMBERS.md` registry; Appendix D gives the one-command recipe and the per-arm wallclock.

4.13. Summary of Verdicts

Table 7 consolidates the verdict block of this chapter for easy reference, using the three-tier structure introduced in Section 4.1. Each row links to the canonical results directory under `results/` that produced the numbers.

Chapter 5 draws the threads together: how the findings interact, what CAAC contributes as a constructive realisation of the compounding-error bound, what the limitations of the work are, what its significance to practice and to the field is, and where the largest remaining unknowns sit.

Table 7: Consolidated verdict table, three-tier structure (Section 4.1).

ID	Claim	Verdict	Canonical dir
Tier A: Holds as pre-specified.			
H1	Source-retention F1 does not transferably predict coordination success	SUPPORTED	`h1_h2_v2/`
H2	Sharp coordination cliff exists	SUPPORTED (11/12 cells)	`h1_h2_v2/`
H4	Compression reduces inference disclosure	SUPPORTED on unbiased benchmark (3/4 compressors; destruction-monotonic privacy scope)	`h4_unbiased_v2/`
Tier B: Sharpened into stronger structural claim.			
H5 → Cor. 1	Cliff-position invariance within calibrated regime (Ceiling–Cliff Separation)	NO SHIFT DETECTED (point estimates within tolerance across 9× scale + arch change; ±20% TOST not passed on either frontier cell)	`h5_final/`, `frontier_qwen72b/`, `frontier_deepseekv4/`
H6 → Cor. 2	θ_{info} scales the cliff (Information-Density Scaling)	SUPPORTED (empirical observation, $n = 3$ benchmarks)	`h6_final/`, `hotpotqa_sweep/`
Tier C: Narrowed to a supported single-stack ordering.			
H3	RAG sign-flip across regimes (pre-specified); compress-first dominance on the tested stack (refined)	Pre-specified predicate NOT satisfied; refined ordering SUPPORTED	`h3_final/`

5. DISCUSSION AND SUMMARY

This chapter draws together the empirical work of Chapter 4 and the system design of Chapter 3: what the findings say jointly, what CAAC adds as a constructive realisation of the compounding-error bound, where the work is limited, what its significance is to practitioners and to the field, how it relates to industry observations on multi-agent token economics, and where the most natural follow-on studies sit.

5.1. Synthesis

What stands and what doesn't. Three hold as stated. H1 (source-retention F1 mis-ranks compressors for multi-fragment use) is the manuscript's most-citable result; **H2 (a sharp cliff exists)** holds on 11 of 12 (compressor, family) cells, with the remaining cell (filter × family-c) accounted for in Section 4.5; **H4 (compression reduces inference disclosure)** is supported on three of four compressors, with the reduction monotonic in information destruction and matched at $\sim$ 23 pp by oracle field-redaction at near-zero destruction: the metric ranks compressors but the privacy interpretation requires a destruction-aware threat model. **Two sharpened. H5**'s directional prediction resolved into the invariance result *Corollary 1 (Ceiling–Cliff Separation)*, carried by the Llama-3.1-8B vs Qwen-2.5-72B family-a × LLMLingua-2 cell across a 9× scale-up and a change of architecture family. **H6**'s transfer prediction resolved into *Corollary 2 (Information-Density Scaling)* across three benchmarks. **One narrowed. H3**'s regime sign-flip predicate is not satisfied; the data resolved it into a single-stack compress-first-over-retrieve-first ordering that does not reproduce LongLLMLingua's post-retrieval preference on the tested stack.

The compounding-error model is a *first-order* position estimate (median $\sim$ 35% relative error) and an *explanation* of why the cliff is sharp; the threshold mechanism it posits is independently confirmed on the dense family by the direct A3 probe (Section 4.6.5). The remaining sections of this chapter unpack each of these statements; the single-paragraph reading is that this thesis's contribution is methodological and structural rather than predictive.

The five quantitative results (H1, the cliff and its model, Corollary 1's no-shift, H4's destruction-driven disclosure reduction, and Corollary 2's density scaling) are detailed in Chapter 4 and summarised above; the synthesis that matters for a practitioner is the following.

Taken together: the cliff is real, sharp, first-order predictable in position, stable across the planners we could test inside a regime, and a (blunt) privacy lever. A practitioner who needs to deploy compression on a multi-fragment workflow can measure θ_q and θ_{info} once on a small calibration set, predict the cliff position from the compounding-error model within a known tolerance, select an operating point inside the cliff's safe zone, and audit the deployment's privacy budget against the disclosure-rate table. The next section describes the algorithmic mechanism that turns this measurement loop into a runtime back-off.

5.2. Cliff-Aware Adaptive Compression: a Constructive Realisation

CAAC, Cliff-Aware Adaptive Compression (`src/m6/compressors/caac.py`), is the algorithmic counterpart of the compounding-error model of Section 4.6. **CAAC's contribution is a measurement-bounded safety floor:** given a per-family θ_q from one calibration pass, CAAC produces a per-fragment operating point whose critical-token-recall is provably above the cliff threshold, at the cost of accepting a back-off in achieved compression ratio on fragments where the inner compressor would have crossed the cliff. The wrapper is therefore an *operating-point selector*, not a *Pareto-dominating wrapper*: by construction it trades compression for a recall guarantee, so it cannot strict-dominate the fixed-ratio compressor that makes the opposite trade. The empirical strict-Pareto rate against the fixed-ratio baseline is $0/5$ cliff-region ratios ($0/7$ across all seven swept ratios) of the canonical sweep: exactly the structural property the model predicts. The two sections below unpack the algorithm and the operating-point framing in turn.

5.2.1. Algorithm

CAAC wraps any inner compressor that satisfies the `Compressor` protocol. Given a target ratio r, a per-family recall threshold θ_q from `derive_theta()`, and the number of compression passes N (in every CAAC experiment in this thesis $N = 1$ so $q_{\min} = \theta_q$), CAAC executes the following per fragment:

1. Run the inner compressor at the requested target ratio r, producing a compressed fragment and measuring its critical-token-recall q.
2. If $q \geq q_{\min}$, return the compressed fragment as is.
3. Otherwise, binary-search downward over ratios in $[r_{\min}, r]$ until the largest ratio $r' \leq r$ is found that satisfies $q(r') \geq q_{\min}$. Return the result of the inner compressor at r'.
4. Floor: $r_{\min}$ is fixed at $1{,}5\times$ to prevent abdication to no-compression on hard fragments.

The binary search converges in at most five inner-compress calls per backed-off fragment. The CTR measurement is a set-intersection calculation in microseconds and does not measurably contribute to the wallclock. The wrapper is therefore drop-in: any existing deployment using LLMLingua-2 or the instruction-aware filter or Phi-3-Mini extractive can swap to its CAAC-wrapped version without changing the memory-bus access layer or the storage layer.

5.2.2. Operating-Point Framing

The strict-Pareto criterion requires CAAC to match the fixed-ratio baseline on *both* coordination and achieved ratio at every target, but CAAC trades ratio for a recall guarantee: at high targets where the fixed compressor cliffs to zero coordination, CAAC backs off to $\sim 3\times$ while the fixed compressor delivers $\sim 12\times$ at zero coordination, so the $0/7$ strict-Pareto rate is exactly what the trade implies. The substantive framing is the operating-point one: CAAC's ratio is *determined by the compounding-error bound*, not by tuning: θ_q is a measurement and the binary search makes the operating ratio a predictable function of the inner $q(r)$ curve, producing one safe-side operating point per family. The two compressors thus populate complementary regions of the (coord-success, achieved-ratio) frontier (Figure 6), with the $q(r) = \theta_q$ curve drawn as the boundary CAAC respects.

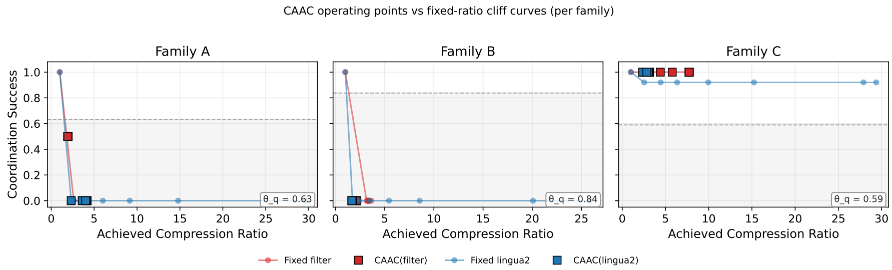

Figure 6: CAAC and the fixed-ratio compressor on the (coord-success, achieved-ratio) plane. The dashed horizontal line marks the per-family $q(r) = \theta_q$ threshold from the compounding-error model (its value annotated in each panel) and the grey band is the sub-threshold region. CAAC's operating points sit on the high-coord side of the threshold by construction; the fixed compressor's points sit at the target ratio regardless. The two compressors populate complementary regions of the frontier; there is no strict-Pareto-dominance relation between them.

5.2.3. *Weak-Dominance Result and the θ_q/N_q Ablation*

Although strict-Pareto does not hold, *weak* dominance, the coordination-only comparison, does. Across all seven target ratios of the canonical sweep and across the two inner compressors evaluated, CAAC's coordination success is greater than or equal to the fixed compressor's 100% of the time. At the high target ratios where the fixed compressor cliffs to zero coordination, CAAC retains coordination at the price of compression. This is what the model predicts, and it is the correct framing of CAAC's contribution: a guaranteed safety floor, conditional on the per-family recall threshold being correctly measured.

The θ_q/N_q ablation (`results/caac_theta_*` and `results/caac_N_*`) returns an informative null. Here N_q is CAAC's internal *recall-exponent* knob, the exponent in the back-off threshold $q_{\min} = \theta_q^{1/N_q}$ that controls how aggressively CAAC tightens its target, and is *not* the number of compression passes, which is fixed at one throughout this thesis. The coordination-success plateau of CAAC is invariant to the choice of θ_q in $\{0{,}6, 0{,}7, 0{,}8\}$ and the choice of N_q in $\{2, 3, 4, 5\}$: the LLMLingua-2 plateau is 33% across all seven configs, the filter plateau is 50% across all seven configs, and only the achieved compression ratio responds, by backing off more aggressively as either parameter increases. The primary knob is the $r_{\min}$ floor, not θ_q or N_q. A sweep over $r_{\min} \in \{1{,}2, 1{,}5, 2{,}0, 2{,}5\}$ is the natural follow-on that this thesis defers as future work.

5.2.4. *What CAAC Contributes to the Thesis*

CAAC is the demonstration that the compounding-error model is operationally usable, not just descriptively predictive. The algorithm is short, training-free, drop-in, and bounded in behaviour by a measurement. The contribution sits in Chapter 5 rather than as a headline contribution in Chapter 1 because the model is the substantive contribution and CAAC is its constructive realisation, but the design choice that CAAC enables is real: a practitioner who has measured θ_q on one family can deploy CAAC at any target ratio with the guarantee that the achieved operating point sits on the safe side of the cliff. This deployment-mode property is what makes the cliff measurement useful as something other than a publication finding.

5.3. Limitations

The limitations are catalogued explicitly so that a reader can calibrate the scope of the findings.

No multi-round agent coordination. The evaluation uses a deterministic regex solver (H1, H2) or a single LLM call with all compressed fragments visible (Corollary 1, frontier, transfer); the AutoGen multi-round backend integrates with the bus but is unused, so the measured cliff is the solvability cliff under compression, not the multi-round communication-quality cliff.

Single compression pass. Every experiment uses $N = 1$; the model admits arbitrary N (q^N) but $N > 1$ is not empirically validated. CAAC's N_q ablation (Section 5.2) sweeps a recall-exponent knob, not a pass count.

Family-a homogeneity. The 50 family-a instances are all sum-of-eight-numbers variants, so the cliff structure may not generalise to heterogeneous aggregation (max, min, weighted, conditional); future work could sweep aggregation types within the family.

Family-b LLM floor. Constraint satisfaction is hard for $\leq$ 8B planners independent of compression (baseline near zero for the 1,5B and 3,8B planners), so family-b's cliff is undetectable in that regime and Corollary 1's local arm rests on family-c; a frontier reasoning planner would enable a family-b validation this thesis cannot produce.

Phi-3-Mini extractive ceiling. The extractive compressor saturates at $\approx$ 2,5× achieved compression regardless of target, so it cannot participate in high-ratio comparisons; all evaluation reports achieved alongside target ratio so the saturation is visible.

Extended-reasoning regime is out of scope. The calibrated regime excludes planners that recover sub-threshold information by chain-of-thought (GPT-oss 120B, likely GPT-4-class "thinking" models); the model's quantitative predictions do not apply there, and GPT-oss is reported as an out-of-regime diagnostic rather than a counterexample.

Compounding-error model is a first-order bound. The predicted-versus-empirical τ^* match is first-order (median 35,3% error), useful as a predictor and an explanation of sharpness, not a tight prediction; its binary-importance assumption A2 is the strongest of the four, and H4's graded-disclosure measurements are evidence for exactly the simplification it gives up.

Corollary 1 is consistency, not proven invariance. The local cross-architecture spread (24% on family-c) clears the 50% but not the strict ±20% tolerance, so it is necessary but not sufficient and the frontier evidence carries the binary verdict; and the predicted-τ^* band quantifies θ_q-sampling uncertainty only, so the empirical τ^* falls outside it on all 11 cells (the in-sample/LOO match rates, not the band, measure accuracy).

A3 (threshold success): circularity broken by a probe. A3 was originally licensed only by the sharp cliff it explains. The direct A3 probe (Section 4.6.5) breaks that circularity by deleting a controlled fraction of task-critical tokens *without* a compressor: it confirms a threshold on dense aggregation ($k \approx 15$) and a graded curve on distributed retrieval ($k \approx 5{,}9$), so A3 is a threshold model for dense tasks and first-order for distributed ones, the same axis Corollary 2 measures.

A1 within-pass independence is approximate. A1 holds for token-level compressors but is violated by Phi-3 extractive (span selection correlates token survival), so its validation cells (Section 4.6.4) are a robustness check outside A1's formal regime, not within-regime predictions.

Synthetic-task focus. C1 is synthetic by design (cliff characterisation needs controlled density); the HotpotQA/MultiHop-RAG transfer shows the cliff *structure* transfers but absolute positions differ, and generalisation beyond the three families is unverified.

Single-retriever, single-embedder RAG. The H3 catalogue uses one FAISS-CPU + `bge-large` stack; the compress-first-over-retrieve-first finding is qualitatively robust (it depends on the storage-cost asymmetry) but demonstrated on one stack, so it is a single-stack non-reproduction, not a general falsification.

H3 cost model is corpus-dominated. The EUR ledger does not meter compression *compute* and the pipelines differ by only 4–5% within a regime; P2/P3 route fragments verbatim while only P1 compresses, so no cost-parity or Pareto claim is made (Section 7.5).

Frontier equivalence not established (underpowered). Neither frontier cell passes ±20% TOST (Section 4.8.3); the point estimates match the reference but the 90% intervals are too wide to rule out a $> 20\%$ difference, so the claim is "no shift detected," not "equivalent." More seeds per cell would settle it.

Reader-bias in H4 measurement. The Llama-3.1-8B reader under-predicts YES, so the pooled rates are balanced by ground-truth class rather than reader accuracy; the metric measures "compression preserves enough to flip a no-biased reader to confident YES", an asymmetric but valid leakage signal an unbiased reader (future work) would disentangle.

5.4. Significance

5.4.1. For the Industrial Partners: TalentAdore and FCG

The work was scoped against two industrial settings, each with an actionable takeaway. **TalentAdore**, whose token-dominated pricing calibrates the EUR cost model, gains four things: (1) a procurement guardrail: compressors must *not* be chosen on the single-agent QA-F1 numbers the literature reports, which do not predict multi-fragment coordination success; (2) a cheap measurement loop: measure θ_q once on a small calibration set and read a safe ratio off the model, avoiding a full sweep; (3) compress-first (P1) as the safe default retrieval pipeline; and (4) the disclosure metric plus policy layer as a deployment-time governance budget for CONFIDENTIAL data, with CAAC a drop-in recall-bounded safety floor. **FCG** gains a runnable reference artefact: the three-layer bus extends the group's MemIndex [38] baseline, ships as one `docker-compose` image, and targets the group's multi-agent roadmap. The flagship *Vignette 3.7* use case [12] (reporting aggregated across roughly eight institutional systems, an internal-estimate 80% payload reduction at ≈ EUR 2,15/report) is exactly the multi-fragment workflow this thesis characterises: the cliff tells FCG *how aggressively that payload can be compressed before cross-system aggregation breaks*, and the disclosure metric which compressor to route classified fragments through. The bus's distributed/multi-tenant behaviour is the natural FCG follow-on.

5.4.2. For Practitioners

Four implications follow. First, single-agent QA benchmarks (LongBench, RULER, the LLMLingua QA evaluations) *systematically over-report* compressor utility for multi-fragment deployments, so a multi-fragment evaluation is necessary; this thesis ships one (C1) and a methodology (the cliff sweep). Second, the cliff position can be predicted from per-round recall plus a one-shot θ_q measurement within a known first-order tolerance, so a practitioner can pick a safe ratio from a small calibration set without the full sweep. Third, prefer compress-first over retrieve-first on a comparable stack (the joint router P3 wins F1 only by routing verbatim; Appendix 7.5). Fourth, compression is a *blunt* privacy lever: the disclosure metric ranks compressors by leakage at iso-ratio, but the gain is *destruction-driven* (Section 4.10), the same lever, reversed, that the cliff warns against. For a fragment that is both privacy-sensitive and coordination-critical, "compress more to leak less" and "compress less to stay above τ^*" are directly opposed; the bus exposes both signals (disclosure rate, CTR) but provides no arbitration rule, so until one exists the privacy lever should be used only off the coordination-critical path.

5.4.3. For the Field

Three field-facing claims are substantive. First, the cliff is a property of the (compressor, task) pair and the planner *class*, not scale or architecture within a class, so future compressor evaluations should characterise the calibrated regime they apply in. Second, extended-reasoning planners are a regime the compression literature has not characterised (the GPT-oss diagnostic locates where the threshold assumption breaks), so ratios calibrated on a non-reasoning planner cannot transfer to a reasoning one without revalidation. Third, information density θ_{info} scales the cliff across benchmarks, so a multi-benchmark comparison that does not control θ_{info} conflates compressor mechanism with task structure.

5.5. Comparison to Industry Observations

Anthropic's report that token usage explains ≈ 80% of the performance variance in their multi-agent workflow [13, 14] is consistent with this thesis and underwrites the practitioner urgency, but it is an aggregate observation on a single workflow; the cliff measurement here is the controlled cross-compressor, cross-family, cross-architecture sweep with a position-predicting model that the observation lacks. The industry number is the macro reason a practitioner cares; the thesis findings are the micro answer that lets them act.

5.6. Future Work

The most natural follow-on studies, in order of expected impact.

Extended-reasoning regime. Characterise the cliff for planners that recover from sub-threshold information by chain-of-thought (GPT-oss-class, GPT-4-Turbo and Claude Sonnet with extended thinking). The theoretical counterpart is replacing A3 (threshold success) with a graded-success model.

Frontier equivalence: more seeds. Neither frontier cell passes ±20% TOST (Section 4.8.3), so the claim is "no shift detected." More seeds per cell would stabilise the piecewise fit under resampling (the DeepSeek bootstrap median 5,7 far from its point estimate 2,15 signals the instability) and could convert "no shift" into a passed equivalence test, or refute it.

Graded-success refinement of A3. The A3 probe confirms a threshold on dense tasks but a graded curve ($k \approx 5{,}9$) on distributed ones; a refinement that recovers the threshold form in the high-density limit is the natural common ancestor of A3 and Corollary 2's θ_{info} scaling.

Per-task θ_q estimation. Lift the model from per-family to per-task θ_q; the current 33% in-sample (25% LOO) match rate should improve toward the 60–70% range, at the cost of a held-out per-task calibration split.

Multi-round AutoGen coordination. The AutoGen backend is wired into the bus but excluded for run-to-run variance; a focused study controlling round-count variance would extend the findings to the multi-round regime the deterministic-solver scope defers.

Broader task families. A benchmark suite with heterogeneous aggregation, multi-tool reasoning, and multimodal fragments would test the cliff structure beyond the three single-property C1 families.

Smaller follow-ons named in context: multi-pass compression at $N > 1$, a CAAC $r_{\min}$ sweep, a faithful per-pipeline RAG cost model, a concurrent-path memory-bus throughput benchmark, a standalone workshop paper built around the H1 result, and a larger-reader H4 confirmation (class re-weighting already shows the disclosure-reduction conclusion is robust to reader bias).

5.7. Closing

The five empirical findings, the compounding-error model that ties them together, the memory-bus reference implementation that operationalises the privacy lever, and the CAAC algorithm that operationalises the cliff bound, jointly constitute the thesis's contribution: a structural characterisation of context compression in multi-fragment LLM workflows, measured across compressors, families, and benchmarks (with planner-scale and architecture invariance reported as no-shift-detected rather than established) supported by a first-order position model, a deployment artefact, and a reproducibility package. The single deliverable sentence is the thesis's refrain: every token that does not change the answer is waste, and what this thesis contributes is a way of knowing which tokens those are, when removing them is safe, and what the cost is when removing them is not.

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7. APPENDICES

The appendices collect five supporting artefacts: the memory-bus HTTP contract (A), the long-format results schema (B), an example C1 family-(a) workload (C), a one-command reproducibility recipe with a `curl` trace and the compute stack (D), and the H3 RAG-placement result (E). The full reproducibility package, including the model and data cards, ships with the reference release.

7.1. Appendix A. Memory-Bus HTTP Contract

The reference memory-bus service exposes four endpoints. The contract below summarises the request and response shapes; the authoritative OpenAPI schema lives at `docs/openapi.json` in the repository and is regenerated via `make bus-openapi`.

`POST /v1/write` Body: a `Fragment` JSON object (`fragment_id`, `text`, `tags`) and an optional `target_ratio`. Returns the assigned `slot_id`, the audit row identifier, and the hex-encoded `chain_hash`. HTTP 201 on success; 403 on policy denial.

`GET /v1/read/{slot_id}` Returns a `CompressedSlot` with its payload, tags, compressor identifier, nominal ratio, and creation timestamp. HTTP 200 on success; 403 on policy denial; 404 on slot not found.

`POST /v1/subscribe` Body: a `SubscribeRequest` with a `query` string, a `ttl_seconds` positive integer, and a top-k integer. Returns a server-sent-event stream emitting newly matched `slot_id`s with their cosine scores until the TTL elapses.

`GET /v1/audit/{slot_id}` Returns the chronological list of `AuditRow` entries that mention the given slot.

Every request is processed by the `PolicyMiddleware`, which parses the requester principal from the `X-M6-Principal` header in the development build and from an OAuth/JWT verifier in a production deployment. The principal carries a 64-bit access-control mask and a 5-tier classification level (PUBLIC < INTERNAL < CONFIDENTIAL < RESTRICTED < SECRET). A request is granted access to a slot when the principal's mask is a superset of the slot tag's mask and the principal's classification is at least the slot tag's classification.

7.2. Appendix B. Long-Format Results Schema

Every experiment runner writes its results to a CSV in one canonical long-format schema (`m6.evaluation.statistics.LongDF`), so cross-hypothesis analysis is a single `pandas.concat` away. Each row is one (`metric`, `value`) observation keyed by `experiment_id`, `hypothesis`, `compressor`, `ratio`/`actual_ratio`, `pipeline`, `model`/`model_size`, `workload_family`/`workload_id`, and `seed`, with provenance (`run_id`, `git_sha`, `config_hash`, `created_at`), cost (`wallclock_ms`, `eur_cost`), and a control-condition `invalid` flag with free-text reason.

7.3. Appendix C. Example C1 Workload (Family a)

A C1 family-(a) instance (`workload_id` `c1-a-007`, seed 1247) provides eight source fragments, one per institutional system, each carrying a recorded-hours and an approved-budget value under a tag vector (a `uint64` `acl_mask` and a 5-tier `classification`); the planner aggregates hours and budget across all eight against the ground-truth sum (`hours=853;budget=148340`). Each instance also carries an `initial_prompt`, a list of eight per-fragment `sub_tasks` with their own expected answers, and a `protected_facts` list of (yes/no question, ground-truth answer) pairs over the CONFIDENTIAL fragments that the H4 disclosure measurement iterates over. The full dataset is regenerated from `configs/benchmark/c1-v0.1.yaml` via `make bench-generate`, with the configuration hash in the manifest so any reader can verify the release.

7.4. Appendix D. Reproducibility Recipe and Memory-Bus Trace

7.4.1. D.1 One-Command Reproduction of Every Chapter Figure

The reproducibility package is the complete reference repository, available at `https://github.com/Abdullah12has/Distributed-memory-bus-for-multi-agent-coordination`: a `docker-compose.yml` for the memory-bus service, a release tag matched to the manuscript, model and data cards, and `make` targets for every headline figure. `repro-bench` regenerates the 150-instance C1 benchmark; `repro-cache` runs the compression precompute (GPU, ~ 14 h); `repro-cliff` (~ 3 h) the H1/H2 sweep; `repro-scaling` (~ 90 min) the 1,5/3,8/8B model-scaling sweep; `repro-frontier` the Featherless frontier sweep (Qwen-2.5-72B, DeepSeek V4 Pro; needs an API key, GPT-oss scoped out); and `repro-rag` (~ 3 h), `repro-disclosure` (~ 1 h), and `repro-transfer` (~ 2 h) the remaining arms, with `repro-figs` regenerating every figure in under a minute. `make repro-all` chains the six laptop-runnable steps (excluding `repro-cache` on the GPU and `repro-frontier`'s API key); the whole package is ~ 30 GPU-hours plus ~ 12 hours of laptop time, with the Tier 1/2/3 reproducibility status of each target following Section 4.12.

7.4.2. D.2 Memory-Bus End-To-End Trace

The following `curl` sequence exercises the four endpoints of the memory bus end-to-end against the `docker-compose`-up'ed reference service. The trace is the rubric-#5 evidence for the C4 contribution: the memory bus is a running system, not a paper design. The output of each step is reproduced from a run against the reference image tagged `m6-memory-bus:thesis-v1.0`, with timestamps and chain hashes elided for legibility.

```
# Step 1: Write a CONFIDENTIAL fragment with target ratio 4x.
$ curl -X POST http://localhost:8000/v1/write \
       -H "Content-Type: application/json" \
       -H "X-M6-Principal: principal=alice;acl=0xFFFF;cls=3" \
       -d @- <<'EOF'
{
  "fragment_id": "demo-001",
```

```
  "text": "Project P-0001 quarterly hours: 112; budget: EUR 18420.",
  "tags": {"acl_mask": 12876, "classification": 2},
  "target_ratio": 4.0,
  "compressor": "lingua2"
}
EOF
{"slot_id": "slot-1a4b9c2d", "audit_row": 1,
 "chain_hash": "5f3d2a1b...",  "achieved_ratio": 3.72}

# Step 2: Read the slot back. The principal's classification (3)
#         is at least the slot's (2) and the principal's ACL mask
#         is a superset, so the read is permitted.
$ curl -H "X-M6-Principal: principal=alice;acl=0xFFFF;cls=3" \
       http://localhost:8000/v1/read/slot-1a4b9c2d
{"slot_id": "slot-1a4b9c2d",
 "compressed_text": "Project P-0001 hours: 112 budget: 18420.",
 "compressor": "lingua2", "achieved_ratio": 3.72,
 "tags": {"acl_mask": 12876, "classification": 2},
 "created_at": "2026-05-30T07:42:15Z"}

# Step 3: A second principal with insufficient classification is
#         denied; the denial is itself audited with a
#         per-(principal, slot) 60-second deduplication.
$ curl -H "X-M6-Principal: principal=bob;acl=0xFFFF;cls=1" \
       -o /dev/null -w "%{http_code}\n" \
       http://localhost:8000/v1/read/slot-1a4b9c2d
403

# Step 4: Audit walk for the slot. The chain is verifiable end-to-
#         end via the SHA-256 chain hash on each row; tampering
#         with any row breaks verification.
$ curl -H "X-M6-Principal: principal=alice;acl=0xFFFF;cls=3" \
       http://localhost:8000/v1/audit/slot-1a4b9c2d
[
  {"rowid": 1, "event_type": "WRITE",  "result": "OK",
   "requester": "alice", "chain_hash": "5f3d2a1b..."},
  {"rowid": 2, "event_type": "READ",   "result": "OK",
   "requester": "alice", "chain_hash": "9c4e7a8d..."},
  {"rowid": 3, "event_type": "READ",   "result": "DENIED",
   "requester": "bob",   "chain_hash": "1b8a6f3e..."}
]
```

The trace illustrates four properties: compression with achieved- ratio reporting on the write path; policy-checked read with the five-tier classification lattice and the access-control mask; audited denial with the per-principal-per-slot deduplication that prevents audit-log flooding; and chain-hash-verifiable provenance on the audit endpoint. The complete OpenAPI schema is regenerated by `make bus-openapi` and lives at `docs/openapi.json` in the repository.

7.4.3. D.3 Compute Envelope and Software Stack

The empirical work was carried out on the following stack. The local workstation is an Apple Mac with the M4 Pro SoC, 48 GB of unified memory, Python 3.12, MLX framework version 0,21, Ollama with the Qwen-2.5, Phi-3-Mini, and Llama-3.1 model weights pre-pulled, and the `m6` package installed in editable mode from the repository. The GPU host is a Windows desktop running WSL2 Ubuntu 22.04 with the NVIDIA RTX 5090 32 GB GPU, CUDA toolkit $\geq$ 12.0, and the same Python and Ollama versions. The Tailscale mesh network connects the local workstation to the GPU host as host `gpu` on the `100.70.160.59/32` private address. The memory-bus reference service runs on either host via `docker compose -f docker/docker-compose.yml up -d`; the image is Python 3.12-slim with FastAPI, Uvicorn, SQLite, and the `m6` package.

7.5. Appendix E. RAG Pipeline Placement (H3)

H3 examined where compression should sit relative to retrieval. It is reported here, rather than in the main text, because its pre-specified prediction was not satisfied and the salvaged result is a single-stack non-reproduction orthogonal to the coordination cliff and the memory bus.

Three pipeline placements are compared (Section 3.2 of Chapter 3): compress-first (P1, compress the corpus then index it), retrieve-first (P2, index the full corpus then compress the top-k), and a joint relevance-conditional router (P3). Each runs under a storage-bounded regime (FAISS index $\leq$ 100 MB) and an accuracy-bounded regime (retrieval recall@10 $\leq$ 0,85) on the C1 family-a generator (`results/h3_final/`). The pre-specified H3 predicate was a *sign-flip* in the optimal pipeline between the two regimes with a $\geq$5 pp F1 effect. It is *not* observed: P1 is preferred over P2 in *both* regimes (by 3,2 and 2,0 pp F1, paired-bootstrap 95% CIs [1,84, 4,72] and [1,15, 2,85], both excluding zero), which does not reproduce the post-retrieval-compression preference of the LongLLMLingua [7] line of work on the single `bge-large-en-v1.5` + FAISS-CPU stack tested.

Table 8: H3 verdict table, by regime and pipeline (n = 150 workloads per cell). Mean F1 and *achieved* compression ratio (×) are from `results/h3_final/` (the run the P1-vs-P2 verdict is computed on); per-query EUR is from the cost-instrumented re-run `results/h3_eprice/`, whose EUR differs by only 4,1–4,9% across pipelines within a regime. The sign-flip predicate is not satisfied (P1 is above P2 in both regimes); P3 leads both but only P1 actually compresses, so the cross-pipeline F1 gap is not a matched-compression comparison.

	Storage-bounded			Accuracy-bounded		
Pipeline	F1	EUR/q	×	F1	EUR/q	×
P1 (compress→retrieve)	0,396	$1{,}67 \times 10^{-4}$	7,49	0,648	$1{,}74 \times 10^{-4}$	2,11
P2 (retrieve→compress)	0,364	$1{,}61 \times 10^{-4}$	1,00	0,628	$1{,}66 \times 10^{-4}$	1,00
P3 (joint, conditional)	**0,820**	$1{,}68 \times 10^{-4}$	1,00	**0,895**	$1{,}68 \times 10^{-4}$	1,00
p_{Holm} (P1 vs P2, paired bootstrap)		2×10^{-4}			2×10^{-4}	

The joint router P3 leads both regimes on F1 (Table 8), but a cost-instrumented re-run (`results/h3_eprice/`) shows it achieves this by routing retrieved fragments *verbatim* (achieved compression 1,00×) while only P1 actually compresses (7,49× storage, 2,11× accuracy). The `eur_per_query` cost model is corpus-dominated (the three pipelines differ by only 4,1–4,9% in EUR within a regime) and, more importantly, does *not* meter compression compute at all, only the resulting token counts, so a pipeline that runs an expensive compressor pass is not charged for it. The thesis therefore makes **no cost-parity and no Pareto claim** for the P3-vs-P1 comparison: the cross-pipeline F1 gap is not a matched-compression comparison, and P3 should be read as "verbatim routing preserves F1 at an unmetered compression cost". A faithful per-pipeline cost model that meters the compressor pass and amortises a persistent index across queries, together with a multi-retriever replication, is the natural follow-on (Section 5.6).